

# NTRK1 fusion-hotspot benchmark report

Study: msk\_impact\_50k\_2026

## Abstract

This report analyzes NTRK1 gene fusions from the msk\_impact\_50k\_2026 study, covering 78 fusion events. 32/44 fusions with known frame status (72.7%) were in-frame. 64/76 fusions (84.2%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ( $p=0.0482628$ , statistically significant at  $\alpha=0.05$ ). No literature benchmark is configured for NTRK1.

## Results summary

### Fusion partner frequency

Fusion-partner frequency was tabulated across 78 analyzed fusion events, identifying 43 distinct fusion partner genes. The most frequent partner was LMNA, observed in 12 events.

### Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 30/32 (93.8%) of in-frame fusions retained the domain,  $p=0.0482628$  (statistically significant at  $\alpha=0.05$ ). The Protein kinase domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.00990099.

### Domain disruption

Domain-disruption analysis was skipped: NTRK1 has no disruption\_required\_domains configured; domain-disruption analysis was skipped.

### Cutpoint detection

Cutpoint detection scanned 76 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 501 aa (exon 12/13 boundary; permutation-corrected  $p=0.019802$ , statistically significant at  $\alpha=0.05$ ). This is -11 aa from the nearest configured domain boundary at 512 aa.

### Corroborating confidence statistics

Corroborating confidence statistics compared Domain\_retention\_flags groups "retained" ( $n=64$ ) and "not\_retained" ( $n=12$ ). For Frame\_status, retained: 30/36 (83.3%, 95% CI 68.1%-92.1%); not\_retained: 2/8 (25.0%, 95% CI 7.1%-59.1%). Welch's t-test comparing Tumor\_variant\_count between groups gave means of 66.3594 vs 43.25,  $p=0.200314$  (not statistically significant at  $\alpha=0.05$ ).

### Composite evidence score

Composite evidence score reported: domain\_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption\_required\_domains not configured for this gene).

## Exon retention

Exon retention produced results for this run (1 row(s) in Exon\_retention); see the accompanying table.

## Expression association

Expression association reported: No mRNA expression data was available for NTRK1 in this cohort; expression-association analysis was skipped.

## Genomic position recurrence

Genomic position recurrence produced results for this run (12 row(s) in genomic\_bin\_recurrence, 7 row(s) in exact\_position\_recurrence, 8 row(s) in protein\_position\_genomic\_spread); see the accompanying table.

## Joint partner

Joint partner reported: NTRK1 has no gene\_pair configured; joint-partner analysis was skipped.

## Mechanistic interpretation

Mechanistic interpretation produced results for this run (2 row(s) in retention\_counter\_intuitive\_events); see the accompanying table.

## Mutation cooccurrence

Mutation cooccurrence reported: NTRK1 has no mutual\_exclusivity\_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

## Window detection

Window detection produced results for this run (163 row(s) in window\_scan, 118 row(s) in top\_windows); see the accompanying table.

# Tables

## composite\_score tables

### composite\_evidence\_ranking

| Partner_gene | Event_count | Recurrence_score    | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                           | Composite_score     | Rank |
|--------------|-------------|---------------------|------------------------|-------------------------|--------------------------|----------------------------|---------------------------------------------------------------------------------|---------------------|------|
| LMNA         | 12          | 0.15384615384615385 | 0.20043213737826426    | None                    | 0.852906976744186        | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.3577818572562792  | 1    |
| TPM3         | 12          | 0.15384615384615385 | 0.20043213737826426    | None                    | 0.7467054263565891       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.3378690665586047  | 2    |
| MEF2D        | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 1.0                      | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.33247718373212887 | 3    |
| TRPM8        | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 1.0                      | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.33247718373212887 | 4    |
| ARHGEF11     | 2           | 0.02564102564102564 | 0.20043213737826426    | None                    | 0.9430232558139535       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.32660173650493746 | 5    |
| RABGAP1L     | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 0.8860465116279069       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.3111109046623614  | 6    |
| VANGL2       | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 0.8860465116279069       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.3111109046623614  | 7    |
| ZBTB7B       | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 0.8860465116279069       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.3111109046623614  | 8    |
| SCP2         | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 0.8651162790697674       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.30718648605771026 | 9    |
| TPR          | 5           | 0.0641025641025641  | 0.20043213737826426    | None                    | 0.7427906976744186       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.30348120877685164 | 10   |
| LTAP1        | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 0.8372093023255813       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.3019539279181754  | 11   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                           | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|---------------------------------------------------------------------------------|---------------------|------|
| ATP1A2       | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | 0.8186046511627907       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.2984655582515216  | 12   |
| PLEKHA6      | 3           | 0.038461538461538464 | 0.20043213737826426    | None                    | 0.7627906976744185       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.297615824161467   | 13   |
| IRF2BP2      | 3           | 0.038461538461538464 | 0.20043213737826426    | None                    | 0.758139534883721        | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.29674373113821123 | 14   |
| ANKRD36      | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | 0.8069767441860465       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.2962853232670126  | 15   |
| RAB25        | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | 0.8046511627906977       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.2958492767553847  | 16   |
| IQGAP3       | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | 0.7976744186046512       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.294541137220501   | 17   |
| SMYD2        | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | 0.7767441860465116       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.2906167186158498  | 18   |
| EML4         | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | 0.7627906976744185       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.28800043954608234 | 19   |
| EPS15        | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | 0.7627906976744185       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.28800043954608234 | 20   |
| F11R         | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | 0.7627906976744185       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.28800043954608234 | 21   |
| PEAR1        | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | 0.7627906976744185       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.28800043954608234 | 22   |
| AFAP1        | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | 0.7488372093023256       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.28538416047631493 | 23   |
| DIAPH1       | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | 0.7488372093023256       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.28538416047631493 | 24   |

| Partner_gene | Event_count | Recurrence_score    | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                           | Composite_score     | Rank |
|--------------|-------------|---------------------|------------------------|-------------------------|--------------------------|----------------------------|---------------------------------------------------------------------------------|---------------------|------|
| TRIM63       | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 0.7488372093023256       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.28538416047631493 | 25   |
| STK11        | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 0.7325581395348837       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.2823318348949196  | 26   |
| KIF21B       | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 0.7255813953488373       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.2810236953600359  | 27   |
| GON4L        | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 0.7162790697674419       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.2792795093135243  | 28   |
| BCAN         | 2           | 0.02564102564102564 | 0.20043213737826426    | None                    | 0.49534883720930234      | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.2426627830165654  | 29   |
| CTRC         | 2           | 0.02564102564102564 | 0.20043213737826426    | None                    | 0.49534883720930234      | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.2426627830165654  | 30   |
| NOS1AP       | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 0.5186046511627906       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.24221555582515214 | 31   |
| NELFCD       | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 0.49534883720930234      | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.2378550907088731  | 32   |
| SEMA4A       | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 0.49534883720930234      | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.2378550907088731  | 33   |
| COP1         | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 0.4534883720930233       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.2300062534995708  | 34   |
| TARS2        | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 0.2697674418604651       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.19555857908096613 | 35   |
| P2RY8        | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | 0.2581395348837209       | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.19337834652282657 | 36   |
| DDR2         | 1           | 0.01282051282051282 | 0.20043213737826426    | None                    | None                     | 0.6202755879498323         | confidence_certainty,<br>domain_retention,<br>recurrence                        | 0.17843345690108176 | 37   |

| Partner_gene | Event_count | Recurrence_score     | Domain_retention_score | Domain_disruption_score | Cutpoint_proximity_score | Confidence_certainty_score | Components_applicable                                                           | Composite_score     | Rank |
|--------------|-------------|----------------------|------------------------|-------------------------|--------------------------|----------------------------|---------------------------------------------------------------------------------|---------------------|------|
| METTL25B     | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | None                     | 0.6202755879498323         | confidence_certainty,<br>domain_retention,<br>recurrence                        | 0.17843345690108176 | 38   |
| SLAMF6       | 3           | 0.038461538461538464 | 0.20043213737826426    | None                    | 0.11395348837209307      | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.17595884741728096 | 39   |
| CADM3        | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | 0.11395348837209307      | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.16634346280189635 | 40   |
| PRCC         | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | 0.11395348837209307      | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.16634346280189635 | 41   |
| TAF2         | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | 0.11395348837209307      | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.16634346280189635 | 42   |
| SHPRH        | 1           | 0.01282051282051282  | 0.20043213737826426    | None                    | 0.0                      | 0.6202755879498323         | confidence_certainty,<br>cutpoint_proximity,<br>domain_retention,<br>recurrence | 0.14497718373212892 | 43   |

## cutpoint\_detection tables

### cutpoint\_scan

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio          | p_value             | neg_log10_p_value   |
|----------|------------------------|------------------|------------------------|------------------|---------------------|---------------------|---------------------|
| 71       | 0                      | 64               | 1                      | 11               | 0.0                 | 0.15789473684210525 | 0.8016323462331666  |
| 120      | 5                      | 59               | 2                      | 10               | 0.423728813559322   | 0.3043106841520683  | 0.5166827995746912  |
| 182      | 5                      | 59               | 3                      | 9                | 0.2542372881355932  | 0.10748982443269033 | 0.9686326464637113  |
| 187      | 6                      | 58               | 3                      | 9                | 0.3103448275862069  | 0.14608214986786255 | 0.83540284831964    |
| 284      | 10                     | 54               | 5                      | 7                | 0.25925925925925924 | 0.05253839246726117 | 1.2795232195834179  |
| 294      | 11                     | 53               | 5                      | 7                | 0.29056603773584905 | 0.11464025310482523 | 0.9406628637664377  |
| 321      | 13                     | 51               | 5                      | 7                | 0.3568627450980392  | 0.14136093824402443 | 0.8496705809825442  |
| 343      | 14                     | 50               | 5                      | 7                | 0.392               | 0.1614921032975089  | 0.7918487091109843  |
| 359      | 15                     | 49               | 5                      | 7                | 0.42857142857142855 | 0.28166616270038797 | 0.550265322823117   |
| 377      | 16                     | 48               | 5                      | 7                | 0.4666666666666667  | 0.29450254706111656 | 0.530910944782467   |
| 379      | 17                     | 47               | 5                      | 7                | 0.5063829787234042  | 0.31247043251825074 | 0.505191071445305   |
| 383      | 18                     | 46               | 5                      | 7                | 0.5478260869565217  | 0.494013437398657   | 0.30626123790080845 |
| 393      | 27                     | 37               | 5                      | 7                | 1.0216216216216216  | 1.0                 | -0.0                |
| 399      | 39                     | 25               | 5                      | 7                | 2.184               | 0.33960073354007925 | 0.46903138034780045 |

| cutpoint | n_positive_at_or_below | n_positive_above | n_negative_at_or_below | n_negative_above | odds_ratio         | p_value               | neg_log10_p_value   |
|----------|------------------------|------------------|------------------------|------------------|--------------------|-----------------------|---------------------|
| 405      | 39                     | 25               | 6                      | 6                | 1.56               | 0.5328211446391313    | 0.27341854881920497 |
| 411      | 40                     | 24               | 6                      | 6                | 1.6666666666666667 | 0.5236156499522178    | 0.28098738162793063 |
| 414      | 41                     | 23               | 6                      | 6                | 1.7826086956521738 | 0.5184932375861125    | 0.2852569034951671  |
| 417      | 41                     | 23               | 7                      | 5                | 1.2732919254658386 | 0.7509447780617822    | 0.12439199836843534 |
| 418      | 48                     | 16               | 7                      | 5                | 2.142857142857143  | 0.29450254706111656   | 0.530910944782467   |
| 423      | 49                     | 15               | 7                      | 5                | 2.3333333333333335 | 0.28166616270038797   | 0.550265322823117   |
| 443      | 50                     | 14               | 7                      | 5                | 2.5510204081632653 | 0.1614921032975089    | 0.7918487091109843  |
| 452      | 61                     | 3                | 8                      | 4                | 10.166666666666666 | 0.010191376831302902  | 1.9917671398557228  |
| 492      | 62                     | 2                | 8                      | 4                | 15.5               | 0.004800736843751964  | 2.3186920995794114  |
| 501      | 64                     | 0                | 9                      | 3                | None               | 0.0031294452347083927 | 2.5045326441976177  |
| 571      | 64                     | 0                | 10                     | 2                | None               | 0.023157894736842103  | 1.6353009244666417  |
| 616      | 64                     | 0                | 11                     | 1                | None               | 0.15789473684210525   | 0.8016323462331666  |

## domain\_retention tables

*frame\_domain\_contingency\_table*

|    |    |
|----|----|
| 30 | 34 |
| 2  | 10 |

*domain\_retention\_descriptives*

| Domain_key | Domain_name           | Quantitative_call_count | Fully_retained_count | Truncated_count | Fully_lost_count | Mean_retained_fraction_among_non_retained_calls |
|------------|-----------------------|-------------------------|----------------------|-----------------|------------------|-------------------------------------------------|
| kinase     | Protein kinase domain | 76                      | 64                   | 3               | 9                | 0.1117283950617284                              |

## exon\_retention tables

*Exon\_retention*

| Exon | Retained_event_count | Total_event_count | Retained_fraction  |
|------|----------------------|-------------------|--------------------|
| 13   | 65                   | 78                | 0.8333333333333334 |

## frequency tables

*Partner\_gene\_counts*

| Partner_gene | Event_count |
|--------------|-------------|
| AFAP1        | 1           |

| Partner_gene | Event_count |
|--------------|-------------|
| ANKRD36      | 1           |
| ARHGEF11     | 2           |
| ATP1A2       | 1           |
| BCAN         | 2           |
| CADM3        | 1           |
| COP1         | 1           |
| CTRC         | 2           |
| DDR2         | 1           |
| DIAPH1       | 1           |
| EML4         | 1           |
| EPS15        | 1           |
| F11R         | 1           |
| GON4L        | 1           |
| IQGAP3       | 1           |
| IRF2BP2      | 3           |
| KIF21B       | 1           |
| LMNA         | 12          |
| LTAP1        | 1           |
| MEF2D        | 1           |
| METTL25B     | 1           |
| NELFCD       | 1           |
| NOS1AP       | 1           |
| P2RY8        | 1           |
| PEAR1        | 1           |
| PLEKHA6      | 3           |
| PRCC         | 1           |
| RAB25        | 1           |
| RABGAP1L     | 1           |
| SCP2         | 1           |
| SEMA4A       | 1           |
| SHPRH        | 1           |
| SLAMF6       | 3           |
| SMYD2        | 1           |
| STK11        | 1           |
| TAF42        | 1           |

| Partner_gene | Event_count |
|--------------|-------------|
| TARS2        | 1           |
| TPM3         | 12          |
| TPR          | 5           |
| TRIM63       | 1           |
| TRPM8        | 1           |
| VANGL2       | 1           |
| ZBTB7B       | 1           |

## genomic\_position\_recurrence tables

### *genomic\_bin\_recurrence*

| chromosome | bin_start | bin_end   | n_events | event_ids                                                                                                                                                     |
|------------|-----------|-----------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1          | 156844000 | 156845000 | 33       | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1, ... (33 total)   |
| 1          | 156845000 | 156846000 | 15       | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24, ... (15 total) |
| 1          | 156843000 | 156844000 | 12       | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0027278-T01-IM6-45,<br>EVT-P-0029326-T01-IM6-10,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0042965-T01-IM6-114, ... (12 total)  |
| 1          | 156836000 | 156837000 | 5        | EVT-P-0024875-T01-IM6-99,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>EVT-P-0055621-T05-IM6-95,<br>EVT-P-0058090-T04-IM7-38                  |
| 1          | 156841000 | 156842000 | 3        | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85                                                                              |
| 1          | 156838000 | 156839000 | 2        | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0069757-T01-IM7-100                                                                                                        |
| 1          | 156831000 | 156832000 | 1        | EVT-P-0036662-T01-IM6-94                                                                                                                                      |
| 1          | 156834000 | 156835000 | 1        | EVT-P-0042526-T02-IM6-92                                                                                                                                      |
| 1          | 156842000 | 156843000 | 1        | EVT-P-0050637-T01-IM6-4                                                                                                                                       |
| 1          | 156846000 | 156847000 | 1        | EVT-P-0025031-T02-IM6-32                                                                                                                                      |
| 1          | 156848000 | 156849000 | 1        | EVT-P-0022936-T01-IM6-98                                                                                                                                      |
| 1          | 156851000 | 156852000 | 1        | EVT-P-0055952-T01-IM6-6                                                                                                                                       |

*exact\_position\_recurrence*

| chromosome | position  | n_events | event_ids                                                                    |
|------------|-----------|----------|------------------------------------------------------------------------------|
| 1          | 156836411 | 3        | EVT-P-0055621-T01-IM6-97, EVT-P-0055621-T02-IM6-96, EVT-P-0055621-T05-IM6-95 |
| 1          | 156844239 | 3        | EVT-P-0029892-T03-IM6-82, EVT-P-0029892-T04-IM6-81, EVT-P-0029892-T05-IM6-80 |
| 1          | 156844525 | 3        | EVT-P-0023994-T01-IM6-31, EVT-P-0023994-T05-IM6-30, EVT-P-0023994-T06-IM6-29 |
| 1          | 156845200 | 3        | EVT-P-0003329-T02-IM6-28, EVT-P-0003329-T03-IM6-26, EVT-P-0003329-T04-IM6-25 |
| 1          | 156843537 | 2        | EVT-P-0049078-T01-IM6-103, EVT-P-0049078-T04-IM6-102                         |
| 1          | 156844046 | 2        | EVT-P-0026375-T01-IM6-106, EVT-P-0026375-T04-IM6-105                         |
| 1          | 156845218 | 2        | EVT-P-0021170-T02-IM6-110, EVT-P-0021170-T03-IM6-109                         |

*protein\_position\_genomic\_spread*

| protein_position_aa | n_events | n_distinct_genomic_positions | genomic_span_bp | spans_multiple_chromosomes |
|---------------------|----------|------------------------------|-----------------|----------------------------|
| 399                 | 12       | 10                           | 78              | False                      |
| 452                 | 12       | 9                            | 280             | False                      |
| 393                 | 9        | 8                            | 318             | False                      |
| 418                 | 7        | 5                            | 224             | False                      |
| 120                 | 6        | 4                            | 1957            | False                      |
| 284                 | 6        | 6                            | 1666            | False                      |
| 501                 | 3        | 3                            | 254             | False                      |
| 321                 | 2        | 1                            | 0               | False                      |

**mechanistic\_interpretation tables**

*retention\_counter\_intuitive\_events*

| event_id                 | sample_id         | partner_gene | breakpoint_protein_position | domain_status |
|--------------------------|-------------------|--------------|-----------------------------|---------------|
| EVT-P-0055952-T01-IM6-6  | P-0055952-T01-IM6 | COP1         | 736                         | disrupted     |
| EVT-P-0024231-T01-IM6-33 | P-0024231-T01-IM6 | MEF2D        | 501                         | lost          |

**window\_detection tables**

*window\_scan*

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value             | neg_log10_p_value  | event_ids_inside                                                                                                                                               |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|---------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| -80      | 120    | 200      | 7               | 69               | 5                 | 59                 | 0.423728813559322  | 0.3043106841520683  | 0.5166827995746912 | EVT-P-0024875-T01-IM6-99,<br>EVT-P-0036662-T01-IM6-94,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>... (7 total) |
| -18      | 182    | 200      | 8               | 68               | 5                 | 59                 | 0.2542372881355932 | 0.10748982443269033 | 0.9686326464637113 | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0024875-T01-IM6-99,<br>EVT-P-0036662-T01-IM6-94,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>... (8 total) |
| -13      | 187    | 200      | 9               | 67               | 6                 | 58                 | 0.3103448275862069 | 0.14608214986786255 | 0.83540284831964   | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0024875-T01-IM6-99,<br>EVT-P-0036662-T01-IM6-94,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>... (9 total) |
| 20       | 120    | 100      | 7               | 69               | 5                 | 59                 | 0.423728813559322  | 0.3043106841520683  | 0.5166827995746912 | EVT-P-0024875-T01-IM6-99,<br>EVT-P-0036662-T01-IM6-94,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>... (7 total) |
| 70       | 120    | 50       | 7               | 69               | 5                 | 59                 | 0.423728813559322  | 0.3043106841520683  | 0.5166827995746912 | EVT-P-0024875-T01-IM6-99,<br>EVT-P-0036662-T01-IM6-94,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>... (7 total) |
| 71       | 121    | 50       | 7               | 69               | 5                 | 59                 | 0.423728813559322  | 0.3043106841520683  | 0.5166827995746912 | EVT-P-0024875-T01-IM6-99,<br>EVT-P-0036662-T01-IM6-94,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>... (7 total) |
| 71       | 171    | 100      | 7               | 69               | 5                 | 59                 | 0.423728813559322  | 0.3043106841520683  | 0.5166827995746912 | EVT-P-0024875-T01-IM6-99,<br>EVT-P-0036662-T01-IM6-94,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>... (7 total) |
| 71       | 271    | 200      | 9               | 67               | 6                 | 58                 | 0.3103448275862069 | 0.14608214986786255 | 0.83540284831964   | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0024875-T01-IM6-99,<br>EVT-P-0036662-T01-IM6-94,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>... (9 total) |
| 82       | 182    | 100      | 7               | 69               | 5                 | 59                 | 0.423728813559322  | 0.3043106841520683  | 0.5166827995746912 | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0024875-T01-IM6-99,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>... (7 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio          | p_value             | neg_log10_p_value   | event_ids_inside                                                                                                                                               |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|---------------------|---------------------|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 84       | 284    | 200      | 14              | 62               | 10                | 54                 | 0.37037037037037035 | 0.21719270795912993 | 0.6631547598671401  | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0024875-T01-IM6-99,<br>EVT-P-0038568-T01-IM6-85,<br>... (14 total)  |
| 87       | 187    | 100      | 8               | 68               | 6                 | 58                 | 0.5172413793103449  | 0.6046542038537166  | 0.21849292335171544 | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0024875-T01-IM6-99,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>... (8 total) |
| 94       | 294    | 200      | 15              | 61               | 11                | 53                 | 0.41509433962264153 | 0.23826873956086597 | 0.6229329325875261  | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0024875-T01-IM6-99,<br>EVT-P-0038568-T01-IM6-85,<br>... (15 total)  |
| 95       | 120    | 25       | 6               | 70               | 5                 | 59                 | 0.9322033898305084  | 1.0                 | -0.0                | EVT-P-0024875-T01-IM6-99,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>EVT-P-0055621-T05-IM6-95,<br>... (6 total) |
| 120      | 145    | 25       | 6               | 70               | 5                 | 59                 | 0.9322033898305084  | 1.0                 | -0.0                | EVT-P-0024875-T01-IM6-99,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>EVT-P-0055621-T05-IM6-95,<br>... (6 total) |
| 120      | 170    | 50       | 6               | 70               | 5                 | 59                 | 0.9322033898305084  | 1.0                 | -0.0                | EVT-P-0024875-T01-IM6-99,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>EVT-P-0055621-T05-IM6-95,<br>... (6 total) |
| 120      | 220    | 100      | 8               | 68               | 6                 | 58                 | 0.5172413793103449  | 0.6046542038537166  | 0.21849292335171544 | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0024875-T01-IM6-99,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>... (8 total) |
| 120      | 320    | 200      | 15              | 61               | 11                | 53                 | 0.41509433962264153 | 0.23826873956086597 | 0.6229329325875261  | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0024875-T01-IM6-99,<br>EVT-P-0038568-T01-IM6-85,<br>... (15 total)  |
| 121      | 321    | 200      | 11              | 65               | 8                 | 56                 | 0.42857142857142855 | 0.365242647320004   | 0.43741851806441334 | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5, ...<br>(11 total)   |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value            | neg_log10_p_value   | event_ids_inside                                                                                                                                              |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|--------------------|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 143      | 343    | 200      | 12              | 64               | 9                 | 55                 | 0.4909090909090909 | 0.3901002779144654 | 0.40882374028888596 | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5, ...<br>(12 total)  |
| 159      | 359    | 200      | 13              | 63               | 10                | 54                 | 0.5555555555555556 | 0.4206035868861286 | 0.3761270279925161  | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5, ...<br>(13 total)  |
| 177      | 377    | 200      | 14              | 62               | 11                | 53                 | 0.6226415094339622 | 0.684604940387127  | 0.16455997114157436 | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5, ...<br>(14 total)  |
| 179      | 379    | 200      | 15              | 61               | 12                | 52                 | 0.6923076923076923 | 0.6947789745681874 | 0.15815333266474796 | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0009855-T01-IM5-13,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>... (15 total) |
| 182      | 382    | 200      | 15              | 61               | 12                | 52                 | 0.6923076923076923 | 0.6947789745681874 | 0.15815333266474796 | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0009855-T01-IM5-13,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>... (15 total) |
| 183      | 383    | 200      | 15              | 61               | 13                | 51                 | 1.2745098039215685 | 1.0                | -0.0                | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0027278-T01-IM6-45,<br>EVT-P-0038568-T01-IM6-85,<br>... (15 total) |
| 184      | 284    | 100      | 7               | 69               | 5                 | 59                 | 0.423728813559322  | 0.3043106841520683 | 0.5166827995746912  | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0050637-T01-IM6-4, ...<br>(7 total)    |
| 187      | 287    | 100      | 7               | 69               | 5                 | 59                 | 0.423728813559322  | 0.3043106841520683 | 0.5166827995746912  | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0050637-T01-IM6-4, ...<br>(7 total)    |
| 187      | 387    | 200      | 15              | 61               | 13                | 51                 | 1.2745098039215685 | 1.0                | -0.0                | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0027278-T01-IM6-45,<br>EVT-P-0038568-T01-IM6-85,<br>... (15 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value              | neg_log10_p_value   | event_ids_inside                                                                                                                                                |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|----------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 193      | 393    | 200      | 23              | 53               | 21                | 43                 | 2.441860465116279  | 0.32681787758256103  | 0.4856941947250095  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8, ...<br>(23 total)   |
| 194      | 294    | 100      | 7               | 69               | 5                 | 59                 | 0.423728813559322  | 0.3043106841520683   | 0.5166827995746912  | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0045731-T01-IM6-35, ...<br>(7 total)     |
| 199      | 399    | 200      | 35              | 41               | 33                | 31                 | 5.32258064516129   | 0.030752567221670444 | 1.5121186235091544  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111, ...<br>(35 total) |
| 205      | 405    | 200      | 36              | 40               | 33                | 31                 | 3.193548387096774  | 0.12020058086869381  | 0.9200934336023401  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111, ...<br>(36 total) |
| 211      | 411    | 200      | 37              | 39               | 34                | 30                 | 3.4                | 0.11495495869977759  | 0.9394722902537481  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111, ...<br>(37 total) |
| 214      | 414    | 200      | 38              | 38               | 35                | 29                 | 3.6206896551724137 | 0.11320514668065682  | 0.9461338282343253  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111, ...<br>(38 total) |
| 217      | 417    | 200      | 39              | 37               | 35                | 29                 | 2.413793103448276  | 0.21731896181834415  | 0.6629023783559742  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111, ...<br>(39 total) |
| 218      | 418    | 200      | 46              | 30               | 42                | 22                 | 3.8181818181818183 | 0.05303061900210486  | 1.275473303531457   | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44, ...<br>(46 total) |
| 221      | 321    | 100      | 9               | 67               | 7                 | 57                 | 0.6140350877192983 | 0.6271831003252879   | 0.20260565238732078 | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0045731-T01-IM6-35, ...<br>(9 total)     |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value              | neg_log10_p_value  | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|----------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 223      | 423    | 200      | 47              | 29               | 43                | 21                 | 4.095238095238095  | 0.04863112187197497  | 1.3130857115712795 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (47 total)  |
| 234      | 284    | 50       | 6               | 70               | 4                 | 60                 | 0.3333333333333333 | 0.238544876303947    | 0.6224299071990397 | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0050637-T01-IM6-4, ...<br>(6 total)       |
| 243      | 343    | 100      | 10              | 66               | 8                 | 56                 | 0.7142857142857143 | 0.6537687167200023   | 0.1845758645463    | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0045731-T01-IM6-35,<br>... (10 total)     |
| 243      | 443    | 200      | 48              | 28               | 44                | 20                 | 4.4                | 0.026102968874188955 | 1.5833100944873653 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (48 total)  |
| 244      | 294    | 50       | 7               | 69               | 5                 | 59                 | 0.423728813559322  | 0.3043106841520683   | 0.5166827995746912 | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0045731-T01-IM6-35,<br>... (7 total)      |
| 252      | 452    | 200      | 60              | 16               | 55                | 9                  | 8.555555555555555  | 0.002229424602155724 | 2.6518072106529775 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (60 total) |
| 259      | 284    | 25       | 6               | 70               | 4                 | 60                 | 0.3333333333333333 | 0.238544876303947    | 0.6224299071990397 | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0050637-T01-IM6-4, ...<br>(6 total)       |
| 259      | 359    | 100      | 11              | 65               | 9                 | 55                 | 0.8181818181818182 | 1.0                  | -0.0               | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0045731-T01-IM6-35,<br>... (11 total)     |
| 269      | 294    | 25       | 7               | 69               | 5                 | 59                 | 0.423728813559322  | 0.3043106841520683   | 0.5166827995746912 | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0045731-T01-IM6-35,<br>... (7 total)      |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value               | neg_log10_p_value   | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|-----------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 271      | 321    | 50       | 9               | 67               | 7                 | 57                 | 0.6140350877192983 | 0.6271831003252879    | 0.20260565238732078 | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0045731-T01-IM6-35,<br>... (9 total)      |
| 277      | 377    | 100      | 12              | 64               | 10                | 54                 | 0.9259259259259259 | 1.0                   | -0.0                | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0042965-T01-IM6-114,<br>... (12 total)    |
| 279      | 379    | 100      | 13              | 63               | 11                | 53                 | 1.0377358490566038 | 1.0                   | -0.0                | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5, ...<br>(13 total)     |
| 283      | 383    | 100      | 14              | 62               | 12                | 52                 | 1.1538461538461537 | 1.0                   | -0.0                | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0027278-T01-IM6-45,<br>EVT-P-0038568-T01-IM6-85,<br>... (14 total)    |
| 284      | 309    | 25       | 7               | 69               | 5                 | 59                 | 0.423728813559322  | 0.3043106841520683    | 0.5166827995746912  | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0045731-T01-IM6-35,<br>... (7 total)      |
| 284      | 334    | 50       | 9               | 67               | 7                 | 57                 | 0.6140350877192983 | 0.6271831003252879    | 0.20260565238732078 | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0045731-T01-IM6-35,<br>... (9 total)      |
| 284      | 384    | 100      | 14              | 62               | 12                | 52                 | 1.1538461538461537 | 1.0                   | -0.0                | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0027278-T01-IM6-45,<br>EVT-P-0038568-T01-IM6-85,<br>... (14 total)    |
| 284      | 484    | 200      | 60              | 16               | 55                | 9                  | 8.555555555555555  | 0.002229424602155724  | 2.6518072106529775  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (60 total) |
| 292      | 492    | 200      | 55              | 21               | 52                | 12                 | 13.0               | 0.0002660917587830724 | 3.574968575920089   | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (55 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio        | p_value               | neg_log10_p_value  | event_ids_inside                                                                                                                                                  |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|-------------------|-----------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 293      | 343    | 50       | 4               | 72               | 4                 | 60                 | None              | 1.0                   | -0.0               | EVT-P-0045731-T01-IM6-35,<br>EVT-P-0049078-T01-IM6-103,<br>EVT-P-0049078-T04-IM6-102,<br>EVT-P-0069266-T01-IM7-112                                                |
| 293      | 393    | 100      | 17              | 59               | 17                | 47                 | None              | 0.057583784217596676  | 1.2396997981099374 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>EVT-P-0021467-T01-IM6-108,<br>... (17 total) |
| 294      | 344    | 50       | 4               | 72               | 4                 | 60                 | None              | 1.0                   | -0.0               | EVT-P-0045731-T01-IM6-35,<br>EVT-P-0049078-T01-IM6-103,<br>EVT-P-0049078-T04-IM6-102,<br>EVT-P-0069266-T01-IM7-112                                                |
| 294      | 394    | 100      | 17              | 59               | 17                | 47                 | None              | 0.057583784217596676  | 1.2396997981099374 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>EVT-P-0021467-T01-IM6-108,<br>... (17 total) |
| 294      | 494    | 200      | 55              | 21               | 52                | 12                 | 13.0              | 0.0002660917587830724 | 3.574968575920089  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (55 total)  |
| 299      | 399    | 100      | 28              | 48               | 28                | 36                 | None              | 0.0027571484021751886 | 2.5595398575753094 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (28 total)   |
| 301      | 501    | 200      | 57              | 19               | 53                | 11                 | 9.636363636363637 | 0.0010544225006145203 | 2.9769853351206437 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (57 total)  |
| 305      | 405    | 100      | 29              | 47               | 28                | 36                 | 8.555555555555555 | 0.023952973648428754  | 1.6206405632958738 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (29 total)   |
| 309      | 359    | 50       | 4               | 72               | 4                 | 60                 | None              | 1.0                   | -0.0               | EVT-P-0049078-T01-IM6-103,<br>EVT-P-0049078-T04-IM6-102,<br>EVT-P-0065034-T02-IM7-115,<br>EVT-P-0069266-T01-IM7-112                                               |
| 311      | 411    | 100      | 30              | 46               | 29                | 35                 | 9.114285714285714 | 0.02224260042534023   | 1.6528144399156985 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (30 total)   |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio        | p_value               | neg_log10_p_value  | event_ids_inside                                                                                                                                                  |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|-------------------|-----------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 314      | 414    | 100      | 31              | 45               | 30                | 34                 | 9.705882352941176 | 0.021834742633238848  | 1.6608519227544536 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (31 total)   |
| 317      | 417    | 100      | 32              | 44               | 30                | 34                 | 4.411764705882353 | 0.06239492128389168   | 1.2048507588455373 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (32 total)   |
| 318      | 418    | 100      | 39              | 37               | 37                | 27                 | 6.851851851851852 | 0.011579953081274666  | 1.936293200244575  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (39 total)   |
| 321      | 371    | 50       | 4               | 72               | 4                 | 60                 | None              | 1.0                   | -0.0               | EVT-P-0049078-T01-IM6-103,<br>EVT-P-0049078-T04-IM6-102,<br>EVT-P-0065034-T02-IM7-115,<br>EVT-P-0069266-T01-IM7-112                                               |
| 321      | 421    | 100      | 39              | 37               | 37                | 27                 | 6.851851851851852 | 0.011579953081274666  | 1.936293200244575  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (39 total)   |
| 321      | 521    | 200      | 57              | 19               | 53                | 11                 | 9.636363636363637 | 0.0010544225006145203 | 2.9769853351206437 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (57 total)  |
| 323      | 423    | 100      | 38              | 38               | 36                | 28                 | 6.428571428571429 | 0.02454800437853655   | 1.6099838079246018 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (38 total)   |
| 329      | 379    | 50       | 4               | 72               | 4                 | 60                 | None              | 1.0                   | -0.0               | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0042965-T01-IM6-114,<br>EVT-P-0065034-T02-IM7-115,<br>EVT-P-0069266-T01-IM7-112                                                |
| 333      | 383    | 50       | 5               | 71               | 5                 | 59                 | None              | 1.0                   | -0.0               | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0027278-T01-IM6-45,<br>EVT-P-0042965-T01-IM6-114,<br>EVT-P-0065034-T02-IM7-115,<br>EVT-P-0069266-T01-IM7-112                   |
| 343      | 393    | 50       | 14              | 62               | 14                | 50                 | None              | 0.10809378796725566   | 0.9661992637623427 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>EVT-P-0021467-T01-IM6-108,<br>... (14 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio        | p_value               | neg_log10_p_value  | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|-------------------|-----------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 343      | 443    | 100      | 39              | 37               | 37                | 27                 | 6.851851851851852 | 0.011579953081274666  | 1.936293200244575  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (39 total)  |
| 343      | 543    | 200      | 55              | 21               | 51                | 13                 | 7.846153846153846 | 0.0025032465666831277 | 2.601496370816874  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (55 total) |
| 349      | 399    | 50       | 25              | 51               | 25                | 39                 | None              | 0.00663070575917867   | 2.178440243691116  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (25 total)  |
| 352      | 452    | 100      | 50              | 26               | 47                | 17                 | 8.294117647058824 | 0.002196626922358729  | 2.6582436978903456 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (50 total) |
| 355      | 405    | 50       | 26              | 50               | 25                | 39                 | 7.051282051282051 | 0.04901447348892081   | 1.309675658171078  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (26 total)  |
| 358      | 383    | 25       | 4               | 72               | 4                 | 60                 | None              | 1.0                   | -0.0               | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0027278-T01-IM6-45,<br>EVT-P-0042965-T01-IM6-114,<br>EVT-P-0065034-T02-IM7-115                                                |
| 359      | 384    | 25       | 4               | 72               | 4                 | 60                 | None              | 1.0                   | -0.0               | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0027278-T01-IM6-45,<br>EVT-P-0042965-T01-IM6-114,<br>EVT-P-0065034-T02-IM7-115                                                |
| 359      | 409    | 50       | 26              | 50               | 25                | 39                 | 7.051282051282051 | 0.04901447348892081   | 1.309675658171078  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (26 total)  |
| 359      | 459    | 100      | 50              | 26               | 47                | 17                 | 8.294117647058824 | 0.002196626922358729  | 2.6582436978903456 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (50 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value               | neg_log10_p_value  | event_ids_inside                                                                                                                                                  |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|-----------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 359      | 559    | 200      | 54              | 22               | 50                | 14                 | 7.142857142857143  | 0.0036886480752847143 | 2.433132777750332  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (54 total)  |
| 361      | 411    | 50       | 26              | 50               | 25                | 39                 | 7.051282051282051  | 0.04901447348892081   | 1.309675658171078  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (26 total)   |
| 364      | 414    | 50       | 27              | 49               | 26                | 38                 | 7.526315789473684  | 0.046619686920376685  | 1.3314306473578994 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (27 total)   |
| 367      | 417    | 50       | 28              | 48               | 26                | 38                 | 3.4210526315789473 | 0.19154920098212158   | 0.7177196552694145 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (28 total)   |
| 368      | 393    | 25       | 12              | 64               | 12                | 52                 | None               | 0.19479032324369738   | 0.7104326217194682 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>EVT-P-0021467-T01-IM6-108,<br>... (12 total) |
| 368      | 418    | 50       | 35              | 41               | 33                | 31                 | 5.32258064516129   | 0.030752567221670444  | 1.5121186235091544 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (35 total)   |
| 371      | 571    | 200      | 54              | 22               | 49                | 15                 | 4.573333333333333  | 0.03213138609039596   | 1.4930705393589638 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (54 total)  |
| 373      | 423    | 50       | 36              | 40               | 34                | 30                 | 5.666666666666667  | 0.0272833549502658    | 1.5641022268747895 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (36 total)   |
| 374      | 399    | 25       | 24              | 52               | 24                | 40                 | None               | 0.01409055149949872   | 1.8510720084164227 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (24 total)   |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio        | p_value               | neg_log10_p_value  | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|-------------------|-----------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 377      | 402    | 25       | 24              | 52               | 24                | 40                 | None              | 0.01409055149949872   | 1.8510720084164227 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (24 total)  |
| 377      | 427    | 50       | 36              | 40               | 34                | 30                 | 5.666666666666667 | 0.0272833549502658    | 1.5641022268747895 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (36 total)  |
| 377      | 477    | 100      | 49              | 27               | 46                | 18                 | 7.666666666666667 | 0.0031244602176755763 | 2.505225000634584  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (49 total) |
| 377      | 577    | 200      | 54              | 22               | 49                | 15                 | 4.573333333333333 | 0.03213138609039596   | 1.4930705393589638 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (54 total) |
| 379      | 404    | 25       | 23              | 53               | 23                | 41                 | None              | 0.013898224210473799  | 1.8570406864264541 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (23 total)  |
| 379      | 429    | 50       | 35              | 41               | 33                | 31                 | 5.32258064516129  | 0.030752567221670444  | 1.5121186235091544 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (35 total)  |
| 379      | 479    | 100      | 48              | 28               | 45                | 19                 | 7.105263157894737 | 0.006608005387249625  | 2.1799296116241407 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (48 total) |
| 379      | 579    | 200      | 53              | 23               | 48                | 16                 | 4.2               | 0.03657634535574214   | 1.4367996906345222 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (53 total) |
| 380      | 405    | 25       | 23              | 53               | 22                | 42                 | 5.761904761904762 | 0.09308913026965836   | 1.0311010272840182 | EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>... (23 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value              | neg_log10_p_value   | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|----------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 383      | 408    | 25       | 23              | 53               | 22                | 42                 | 5.761904761904762  | 0.09308913026965836  | 1.0311010272840182  | EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>... (23 total) |
| 383      | 433    | 50       | 34              | 42               | 32                | 32                 | 5.0                | 0.055111739359058345 | 1.2587558821677678  | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1, ...<br>(34 total)   |
| 383      | 483    | 100      | 47              | 29               | 44                | 20                 | 6.6                | 0.007671143250306442 | 2.115139907198372   | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (47 total) |
| 383      | 583    | 200      | 52              | 24               | 47                | 17                 | 3.8705882352941177 | 0.04308632903484734  | 1.3656605063252698  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (52 total) |
| 386      | 411    | 25       | 23              | 53               | 22                | 42                 | 5.761904761904762  | 0.09308913026965836  | 1.0311010272840182  | EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>... (23 total) |
| 389      | 414    | 25       | 24              | 52               | 23                | 41                 | 6.170731707317073  | 0.08981716348376248  | 1.0466406645126898  | EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>... (24 total) |
| 392      | 417    | 25       | 25              | 51               | 23                | 41                 | 2.8048780487804876 | 0.3165037413751296   | 0.49962115184261413 | EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>... (25 total) |
| 392      | 492    | 100      | 47              | 29               | 44                | 20                 | 6.6                | 0.007671143250306442 | 2.115139907198372   | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (47 total) |
| 393      | 418    | 25       | 32              | 44               | 30                | 34                 | 4.411764705882353  | 0.06239492128389168  | 1.2048507588455373  | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1, ...<br>(32 total)   |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value              | neg_log10_p_value   | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|----------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 393      | 443    | 50       | 34              | 42               | 32                | 32                 | 5.0                | 0.055111739359058345 | 1.2587558821677678  | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1, ...<br>(34 total)   |
| 393      | 493    | 100      | 47              | 29               | 44                | 20                 | 6.6                | 0.007671143250306442 | 2.115139907198372   | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (47 total) |
| 393      | 593    | 200      | 51              | 25               | 46                | 18                 | 3.577777777777778  | 0.0517429931906439   | 1.2861484521447824  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (51 total) |
| 398      | 423    | 25       | 24              | 52               | 22                | 42                 | 2.619047619047619  | 0.318895716002397    | 0.4963513149653078  | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0021262-T01-IM6-41,<br>... (24 total)  |
| 399      | 424    | 25       | 24              | 52               | 22                | 42                 | 2.619047619047619  | 0.318895716002397    | 0.4963513149653078  | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0021262-T01-IM6-41,<br>... (24 total)  |
| 399      | 449    | 50       | 25              | 51               | 23                | 41                 | 2.8048780487804876 | 0.3165037413751296   | 0.49962115184261413 | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0021262-T01-IM6-41,<br>... (25 total)  |
| 399      | 499    | 100      | 38              | 38               | 35                | 29                 | 3.6206896551724137 | 0.11320514668065682  | 0.9461338282343253  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (38 total) |
| 399      | 599    | 200      | 42              | 34               | 37                | 27                 | 1.9185185185185185 | 0.3541470110878887   | 0.4508164191868243  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (42 total) |
| 401      | 501    | 100      | 29              | 47               | 25                | 39                 | 1.2820512820512822 | 1.0                  | -0.0                | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (29 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value            | neg_log10_p_value   | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|--------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 402      | 452    | 50       | 25              | 51               | 22                | 42                 | 1.5714285714285714 | 0.740344859540053  | 0.1305659347182275  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (25 total) |
| 405      | 430    | 25       | 12              | 64               | 10                | 54                 | 0.9259259259259259 | 1.0                | -0.0                | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0022925-T01-IM6-16,<br>EVT-P-0023994-T01-IM6-31,<br>EVT-P-0023994-T05-IM6-30,<br>... (12 total)  |
| 405      | 455    | 50       | 25              | 51               | 22                | 42                 | 1.5714285714285714 | 0.740344859540053  | 0.1305659347182275  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (25 total) |
| 405      | 505    | 100      | 29              | 47               | 25                | 39                 | 1.2820512820512822 | 1.0                | -0.0                | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (29 total) |
| 405      | 605    | 200      | 30              | 46               | 25                | 39                 | 0.8974358974358975 | 1.0                | -0.0                | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (30 total) |
| 411      | 436    | 25       | 11              | 65               | 10                | 54                 | 2.037037037037037  | 1.0                | -0.0                | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0022925-T01-IM6-16,<br>EVT-P-0023994-T01-IM6-31,<br>EVT-P-0023994-T05-IM6-30,<br>... (11 total)  |
| 411      | 461    | 50       | 24              | 52               | 22                | 42                 | 2.619047619047619  | 0.318895716002397  | 0.4963513149653078  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (24 total) |
| 411      | 511    | 100      | 28              | 48               | 25                | 39                 | 1.9230769230769231 | 0.5176833019050124 | 0.28593584314537324 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (28 total) |
| 411      | 611    | 200      | 29              | 47               | 25                | 39                 | 1.2820512820512822 | 1.0                | -0.0                | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (29 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value             | neg_log10_p_value   | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|---------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 414      | 439    | 25       | 10              | 66               | 9                 | 55                 | 1.8                | 1.0                 | -0.0                | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0023994-T01-IM6-31,<br>EVT-P-0023994-T05-IM6-30,<br>EVT-P-0023994-T06-IM6-29,<br>... (10 total)  |
| 414      | 464    | 50       | 23              | 53               | 21                | 43                 | 2.441860465116279  | 0.32681787758256103 | 0.4856941947250095  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (23 total) |
| 414      | 514    | 100      | 27              | 49               | 24                | 40                 | 1.8                | 0.5213280124699622  | 0.2828889385407058  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (27 total) |
| 414      | 614    | 200      | 28              | 48               | 24                | 40                 | 1.2                | 1.0                 | -0.0                | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (28 total) |
| 416      | 616    | 200      | 28              | 48               | 23                | 41                 | 0.7853658536585366 | 0.7509447780617824  | 0.1243919983684352  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (28 total) |
| 417      | 442    | 25       | 9               | 67               | 8                 | 56                 | 1.5714285714285714 | 1.0                 | -0.0                | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0023994-T01-IM6-31,<br>EVT-P-0023994-T05-IM6-30,<br>EVT-P-0023994-T06-IM6-29,<br>... (9 total)   |
| 417      | 467    | 50       | 22              | 54               | 20                | 44                 | 2.272727272727273  | 0.4906657002169846  | 0.3092143001343047  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (22 total) |
| 417      | 517    | 100      | 26              | 50               | 23                | 41                 | 1.6829268292682926 | 0.5294734121362175  | 0.27615584339868837 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (26 total) |
| 417      | 617    | 200      | 28              | 48               | 23                | 41                 | 0.7853658536585366 | 0.7509447780617824  | 0.1243919983684352  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (28 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value             | neg_log10_p_value   | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|---------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 418      | 443    | 25       | 9               | 67               | 9                 | 55                 | None               | 0.3393946163658613  | 0.4692950509433365  | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0023994-T01-IM6-31,<br>EVT-P-0023994-T05-IM6-30,<br>EVT-P-0023994-T06-IM6-29,<br>... (9 total)   |
| 418      | 468    | 50       | 21              | 55               | 20                | 44                 | 5.0                | 0.16139183599106333 | 0.7921184378525271  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (21 total) |
| 418      | 518    | 100      | 25              | 51               | 23                | 41                 | 2.8048780487804876 | 0.3165037413751296  | 0.49962115184261413 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (25 total) |
| 418      | 618    | 200      | 27              | 49               | 23                | 41                 | 1.1219512195121952 | 1.0                 | -0.0                | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (27 total) |
| 423      | 473    | 50       | 14              | 62               | 13                | 51                 | 2.803921568627451  | 0.4465709331321285  | 0.35010954815025525 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (14 total) |
| 423      | 523    | 100      | 18              | 58               | 16                | 48                 | 1.6666666666666667 | 0.7198940138934029  | 0.14273143769289975 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (18 total) |
| 423      | 623    | 200      | 20              | 56               | 16                | 48                 | 0.6666666666666666 | 0.7215985127946598  | 0.14170437054058568 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (20 total) |
| 427      | 452    | 25       | 13              | 63               | 12                | 52                 | 2.5384615384615383 | 0.6786540902057518  | 0.168351529097454   | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (13 total) |
| 442      | 492    | 50       | 14              | 62               | 13                | 51                 | 2.803921568627451  | 0.4465709331321285  | 0.35010954815025525 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (14 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value             | neg_log10_p_value   | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|---------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 443      | 468    | 25       | 13              | 63               | 12                | 52                 | 2.5384615384615383 | 0.6786540902057518  | 0.168351529097454   | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (13 total) |
| 443      | 493    | 50       | 14              | 62               | 13                | 51                 | 2.803921568627451  | 0.4465709331321285  | 0.35010954815025525 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (14 total) |
| 443      | 543    | 100      | 17              | 59               | 15                | 49                 | 1.530612244897959  | 1.0                 | -0.0                | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (17 total) |
| 443      | 643    | 200      | 19              | 57               | 15                | 49                 | 0.6122448979591837 | 0.48095522729154383 | 0.3178953507472748  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (19 total) |
| 451      | 501    | 50       | 16              | 60               | 14                | 50                 | 1.4                | 1.0                 | -0.0                | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (16 total) |
| 452      | 477    | 25       | 12              | 64               | 11                | 53                 | 2.2830188679245285 | 0.6777385747932327  | 0.16893779495157404 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (12 total) |
| 452      | 502    | 50       | 16              | 60               | 14                | 50                 | 1.4                | 1.0                 | -0.0                | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (16 total) |
| 452      | 552    | 100      | 16              | 60               | 14                | 50                 | 1.4                | 1.0                 | -0.0                | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (16 total) |
| 452      | 652    | 200      | 18              | 58               | 14                | 50                 | 0.56               | 0.46254663915796645 | 0.33484447025957326 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (18 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio          | p_value              | neg_log10_p_value  | event_ids_inside                                                                                                                                               |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|---------------------|----------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 471      | 571    | 100      | 5               | 71               | 3                 | 61                 | 0.2459016393442623  | 0.17460589645160657  | 0.757941094232076  | EVT-P-0003705-T01-IM5-24,<br>EVT-P-0007755-T02-IM5-117,<br>EVT-P-0017558-T02-IM7-3,<br>EVT-P-0024231-T01-IM6-33,<br>EVT-P-0025031-T02-IM6-32                   |
| 476      | 501    | 25       | 4               | 72               | 3                 | 61                 | 0.5409836065573771  | 0.504763537870964    | 0.2969120243529968 | EVT-P-0003705-T01-IM5-24,<br>EVT-P-0007755-T02-IM5-117,<br>EVT-P-0017558-T02-IM7-3,<br>EVT-P-0024231-T01-IM6-33                                                |
| 492      | 517    | 25       | 4               | 72               | 3                 | 61                 | 0.5409836065573771  | 0.504763537870964    | 0.2969120243529968 | EVT-P-0003705-T01-IM5-24,<br>EVT-P-0007755-T02-IM5-117,<br>EVT-P-0017558-T02-IM7-3,<br>EVT-P-0024231-T01-IM6-33                                                |
| 492      | 542    | 50       | 4               | 72               | 3                 | 61                 | 0.5409836065573771  | 0.504763537870964    | 0.2969120243529968 | EVT-P-0003705-T01-IM5-24,<br>EVT-P-0007755-T02-IM5-117,<br>EVT-P-0017558-T02-IM7-3,<br>EVT-P-0024231-T01-IM6-33                                                |
| 492      | 592    | 100      | 5               | 71               | 3                 | 61                 | 0.2459016393442623  | 0.17460589645160657  | 0.757941094232076  | EVT-P-0003705-T01-IM5-24,<br>EVT-P-0007755-T02-IM5-117,<br>EVT-P-0017558-T02-IM7-3,<br>EVT-P-0024231-T01-IM6-33,<br>EVT-P-0025031-T02-IM6-32                   |
| 492      | 692    | 200      | 6               | 70               | 3                 | 61                 | 0.14754098360655737 | 0.046727936746925956 | 1.330423394648854  | EVT-P-0003705-T01-IM5-24,<br>EVT-P-0007755-T02-IM5-117,<br>EVT-P-0017558-T02-IM7-3,<br>EVT-P-0022936-T01-IM6-98,<br>EVT-P-0024231-T01-IM6-33,<br>... (6 total) |
| 501      | 601    | 100      | 4               | 72               | 2                 | 62                 | 0.16129032258064516 | 0.11506927258909953  | 0.939040632291921  | EVT-P-0007755-T02-IM5-117,<br>EVT-P-0017558-T02-IM7-3,<br>EVT-P-0024231-T01-IM6-33,<br>EVT-P-0025031-T02-IM6-32                                                |
| 501      | 701    | 200      | 5               | 71               | 2                 | 62                 | 0.0967741935483871  | 0.02576433679533895  | 1.5889810323153342 | EVT-P-0007755-T02-IM5-117,<br>EVT-P-0017558-T02-IM7-3,<br>EVT-P-0022936-T01-IM6-98,<br>EVT-P-0024231-T01-IM6-33,<br>EVT-P-0025031-T02-IM6-32                   |

### top\_windows

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio        | p_value               | neg_log10_p_value  | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|-------------------|-----------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 292      | 492    | 200      | 55              | 21               | 52                | 12                 | 13.0              | 0.0002660917587830724 | 3.574968575920089  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (55 total) |
| 301      | 501    | 200      | 57              | 19               | 53                | 11                 | 9.636363636363637 | 0.0010544225006145203 | 2.9769853351206437 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (57 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio        | p_value                | neg_log10_p_value  | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|-------------------|------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 352      | 452    | 100      | 50              | 26               | 47                | 17                 | 8.294117647058824 | 0.002196626922358729   | 2.6582436978903456 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (50 total) |
| 252      | 452    | 200      | 60              | 16               | 55                | 9                  | 8.555555555555555 | 0.002229424602155724   | 2.6518072106529775 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (60 total) |
| 343      | 543    | 200      | 55              | 21               | 51                | 13                 | 7.846153846153846 | 0.0025032465666831277  | 2.601496370816874  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (55 total) |
| 299      | 399    | 100      | 28              | 48               | 28                | 36                 | None              | 0.0027571484021751886  | 2.5595398575753094 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (28 total)  |
| 377      | 477    | 100      | 49              | 27               | 46                | 18                 | 7.666666666666667 | 0.00312444602176755763 | 2.505225000634584  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (49 total) |
| 359      | 559    | 200      | 54              | 22               | 50                | 14                 | 7.142857142857143 | 0.0036886480752847143  | 2.433132777750332  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (54 total) |
| 379      | 479    | 100      | 48              | 28               | 45                | 19                 | 7.105263157894737 | 0.006608005387249625   | 2.1799296116241407 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (48 total) |
| 349      | 399    | 50       | 25              | 51               | 25                | 39                 | None              | 0.00663070575917867    | 2.178440243691116  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (25 total)  |
| 383      | 483    | 100      | 47              | 29               | 44                | 20                 | 6.6               | 0.007671143250306442   | 2.115139907198372  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (47 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio        | p_value              | neg_log10_p_value  | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|-------------------|----------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 392      | 492    | 100      | 47              | 29               | 44                | 20                 | 6.6               | 0.007671143250306442 | 2.115139907198372  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (47 total) |
| 318      | 418    | 100      | 39              | 37               | 37                | 27                 | 6.851851851851852 | 0.011579953081274666 | 1.936293200244575  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (39 total)  |
| 343      | 443    | 100      | 39              | 37               | 37                | 27                 | 6.851851851851852 | 0.011579953081274666 | 1.936293200244575  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (39 total)  |
| 379      | 404    | 25       | 23              | 53               | 23                | 41                 | None              | 0.013898224210473799 | 1.8570406864264541 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (23 total)  |
| 374      | 399    | 25       | 24              | 52               | 24                | 40                 | None              | 0.01409055149949872  | 1.8510720084164227 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (24 total)  |
| 314      | 414    | 100      | 31              | 45               | 30                | 34                 | 9.705882352941176 | 0.021834742633238848 | 1.6608519227544536 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (31 total)  |
| 311      | 411    | 100      | 30              | 46               | 29                | 35                 | 9.114285714285714 | 0.02224260042534023  | 1.6528144399156985 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (30 total)  |
| 305      | 405    | 100      | 29              | 47               | 28                | 36                 | 8.555555555555555 | 0.023952973648428754 | 1.6206405632958738 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (29 total)  |
| 323      | 423    | 100      | 38              | 38               | 36                | 28                 | 6.428571428571429 | 0.02454800437853655  | 1.6099838079246018 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (38 total)  |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value              | neg_log10_p_value  | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|----------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 501      | 701    | 200      | 5               | 71               | 2                 | 62                 | 0.0967741935483871 | 0.02576433679533895  | 1.5889810323153342 | EVT-P-0007755-T02-IM5-117,<br>EVT-P-0017558-T02-IM7-3,<br>EVT-P-0022936-T01-IM6-98,<br>EVT-P-0024231-T01-IM6-33,<br>EVT-P-0025031-T02-IM6-32                     |
| 243      | 443    | 200      | 48              | 28               | 44                | 20                 | 4.4                | 0.026102968874188955 | 1.5833100944873653 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (48 total)  |
| 373      | 423    | 50       | 36              | 40               | 34                | 30                 | 5.666666666666667  | 0.0272833549502658   | 1.5641022268747895 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (36 total)  |
| 368      | 418    | 50       | 35              | 41               | 33                | 31                 | 5.32258064516129   | 0.030752567221670444 | 1.5121186235091544 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (35 total)  |
| 379      | 429    | 50       | 35              | 41               | 33                | 31                 | 5.32258064516129   | 0.030752567221670444 | 1.5121186235091544 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (35 total)  |
| 199      | 399    | 200      | 35              | 41               | 33                | 31                 | 5.32258064516129   | 0.030752567221670444 | 1.5121186235091544 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (35 total)  |
| 371      | 571    | 200      | 54              | 22               | 49                | 15                 | 4.573333333333333  | 0.03213138609039596  | 1.4930705393589638 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (54 total) |
| 379      | 579    | 200      | 53              | 23               | 48                | 16                 | 4.2                | 0.03657634535574214  | 1.4367996906345222 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (53 total) |
| 383      | 583    | 200      | 52              | 24               | 47                | 17                 | 3.8705882352941177 | 0.04308632903484734  | 1.3656605063252698 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (52 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio          | p_value              | neg_log10_p_value  | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|---------------------|----------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 364      | 414    | 50       | 27              | 49               | 26                | 38                 | 7.526315789473684   | 0.046619686920376685 | 1.3314306473578994 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (27 total)  |
| 492      | 692    | 200      | 6               | 70               | 3                 | 61                 | 0.14754098360655737 | 0.046727936746925956 | 1.330423394648854  | EVT-P-0003705-T01-IM5-24,<br>EVT-P-0007755-T02-IM5-117,<br>EVT-P-0017558-T02-IM7-3,<br>EVT-P-0022936-T01-IM6-98,<br>EVT-P-0024231-T01-IM6-33,<br>... (6 total)   |
| 223      | 423    | 200      | 47              | 29               | 43                | 21                 | 4.095238095238095   | 0.04863112187197497  | 1.3130857115712795 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (47 total)  |
| 355      | 405    | 50       | 26              | 50               | 25                | 39                 | 7.051282051282051   | 0.04901447348892081  | 1.309675658171078  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (26 total)  |
| 361      | 411    | 50       | 26              | 50               | 25                | 39                 | 7.051282051282051   | 0.04901447348892081  | 1.309675658171078  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (26 total)  |
| 393      | 593    | 200      | 51              | 25               | 46                | 18                 | 3.5777777777777778  | 0.0517429931906439   | 1.2861484521447824 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (51 total) |
| 218      | 418    | 200      | 46              | 30               | 42                | 22                 | 3.8181818181818183  | 0.05303061900210486  | 1.275473303531457  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>... (46 total)  |
| 383      | 433    | 50       | 34              | 42               | 32                | 32                 | 5.0                 | 0.055111739359058345 | 1.2587558821677678 | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1, ...<br>(34 total)   |
| 393      | 443    | 50       | 34              | 42               | 32                | 32                 | 5.0                 | 0.055111739359058345 | 1.2587558821677678 | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1, ...<br>(34 total)   |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value              | neg_log10_p_value  | event_ids_inside                                                                                                                                                  |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|----------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 293      | 393    | 100      | 17              | 59               | 17                | 47                 | None               | 0.057583784217596676 | 1.2396997981099374 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>EVT-P-0021467-T01-IM6-108,<br>... (17 total) |
| 393      | 418    | 25       | 32              | 44               | 30                | 34                 | 4.411764705882353  | 0.06239492128389168  | 1.2048507588455373 | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1, ...<br>(32 total)    |
| 317      | 417    | 100      | 32              | 44               | 30                | 34                 | 4.411764705882353  | 0.06239492128389168  | 1.2048507588455373 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (32 total)   |
| 389      | 414    | 25       | 24              | 52               | 23                | 41                 | 6.170731707317073  | 0.08981716348376248  | 1.0466406645126898 | EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>... (24 total)  |
| 380      | 405    | 25       | 23              | 53               | 22                | 42                 | 5.761904761904762  | 0.09308913026965836  | 1.0311010272840182 | EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>... (23 total)  |
| 386      | 411    | 25       | 23              | 53               | 22                | 42                 | 5.761904761904762  | 0.09308913026965836  | 1.0311010272840182 | EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>... (23 total)  |
| -18      | 182    | 200      | 8               | 68               | 5                 | 59                 | 0.2542372881355932 | 0.10748982443269033  | 0.9686326464637113 | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0024875-T01-IM6-99,<br>EVT-P-0036662-T01-IM6-94,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>... (8 total)    |
| 343      | 393    | 50       | 14              | 62               | 14                | 50                 | None               | 0.10809378796725566  | 0.9661992637623427 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>EVT-P-0021467-T01-IM6-108,<br>... (14 total) |
| 399      | 499    | 100      | 38              | 38               | 35                | 29                 | 3.6206896551724137 | 0.11320514668065682  | 0.9461338282343253 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (38 total)  |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio          | p_value             | neg_log10_p_value  | event_ids_inside                                                                                                                                                  |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|---------------------|---------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 214      | 414    | 200      | 38              | 38               | 35                | 29                 | 3.6206896551724137  | 0.11320514668065682 | 0.9461338282343253 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (38 total)   |
| 211      | 411    | 200      | 37              | 39               | 34                | 30                 | 3.4                 | 0.11495495869977759 | 0.9394722902537481 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (37 total)   |
| 501      | 601    | 100      | 4               | 72               | 2                 | 62                 | 0.16129032258064516 | 0.11506927258909953 | 0.939040632291921  | EVT-P-0007755-T02-IM5-117,<br>EVT-P-0017558-T02-IM7-3,<br>EVT-P-0024231-T01-IM6-33,<br>EVT-P-0025031-T02-IM6-32                                                   |
| 205      | 405    | 200      | 36              | 40               | 33                | 31                 | 3.193548387096774   | 0.12020058086869381 | 0.9200934336023401 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (36 total)   |
| -13      | 187    | 200      | 9               | 67               | 6                 | 58                 | 0.3103448275862069  | 0.14608214986786255 | 0.83540284831964   | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0024875-T01-IM6-99,<br>EVT-P-0036662-T01-IM6-94,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>... (9 total)    |
| 418      | 468    | 50       | 21              | 55               | 20                | 44                 | 5.0                 | 0.16139183599106333 | 0.7921184378525271 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (21 total)  |
| 471      | 571    | 100      | 5               | 71               | 3                 | 61                 | 0.2459016393442623  | 0.17460589645160657 | 0.757941094232076  | EVT-P-0003705-T01-IM5-24,<br>EVT-P-0007755-T02-IM5-117,<br>EVT-P-0017558-T02-IM7-3,<br>EVT-P-0024231-T01-IM6-33,<br>EVT-P-0025031-T02-IM6-32                      |
| 367      | 417    | 50       | 28              | 48               | 26                | 38                 | 3.4210526315789473  | 0.19154920098212158 | 0.7177196552694145 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (28 total)   |
| 368      | 393    | 25       | 12              | 64               | 12                | 52                 | None                | 0.19479032324369738 | 0.7104326217194682 | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>EVT-P-0021467-T01-IM6-108,<br>... (12 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio          | p_value             | neg_log10_p_value   | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|---------------------|---------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 84       | 284    | 200      | 14              | 62               | 10                | 54                 | 0.37037037037037035 | 0.21719270795912993 | 0.6631547598671401  | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0024875-T01-IM6-99,<br>EVT-P-0038568-T01-IM6-85,<br>... (14 total)    |
| 217      | 417    | 200      | 39              | 37               | 35                | 29                 | 2.413793103448276   | 0.21731896181834415 | 0.6629023783559742  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>... (39 total)  |
| 94       | 294    | 200      | 15              | 61               | 11                | 53                 | 0.41509433962264153 | 0.23826873956086597 | 0.6229329325875261  | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0024875-T01-IM6-99,<br>EVT-P-0038568-T01-IM6-85,<br>... (15 total)    |
| 259      | 284    | 25       | 6               | 70               | 4                 | 60                 | 0.3333333333333333  | 0.238544876303947   | 0.6224299071990397  | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0050637-T01-IM6-4, ...<br>(6 total)       |
| 269      | 294    | 25       | 7               | 69               | 5                 | 59                 | 0.423728813559322   | 0.3043106841520683  | 0.5166827995746912  | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0045731-T01-IM6-35,<br>... (7 total)      |
| 70       | 120    | 50       | 7               | 69               | 5                 | 59                 | 0.423728813559322   | 0.3043106841520683  | 0.5166827995746912  | EVT-P-0024875-T01-IM6-99,<br>EVT-P-0036662-T01-IM6-94,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>... (7 total)   |
| 82       | 182    | 100      | 7               | 69               | 5                 | 59                 | 0.423728813559322   | 0.3043106841520683  | 0.5166827995746912  | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0024875-T01-IM6-99,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>... (7 total)   |
| 184      | 284    | 100      | 7               | 69               | 5                 | 59                 | 0.423728813559322   | 0.3043106841520683  | 0.5166827995746912  | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0050637-T01-IM6-4, ...<br>(7 total)       |
| 392      | 417    | 25       | 25              | 51               | 23                | 41                 | 2.8048780487804876  | 0.3165037413751296  | 0.49962115184261413 | EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0016347-T01-IM6-116,<br>... (25 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio          | p_value             | neg_log10_p_value   | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|---------------------|---------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 399      | 449    | 50       | 25              | 51               | 23                | 41                 | 2.8048780487804876  | 0.3165037413751296  | 0.49962115184261413 | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0021262-T01-IM6-41,<br>... (25 total)  |
| 418      | 518    | 100      | 25              | 51               | 23                | 41                 | 2.8048780487804876  | 0.3165037413751296  | 0.49962115184261413 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (25 total) |
| 398      | 423    | 25       | 24              | 52               | 22                | 42                 | 2.619047619047619   | 0.318895716002397   | 0.4963513149653078  | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0010628-T01-IM5-43,<br>EVT-P-0010628-T03-IM6-44,<br>EVT-P-0021262-T01-IM6-41,<br>... (24 total)  |
| 411      | 461    | 50       | 24              | 52               | 22                | 42                 | 2.619047619047619   | 0.318895716002397   | 0.4963513149653078  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (24 total) |
| 414      | 464    | 50       | 23              | 53               | 21                | 43                 | 2.441860465116279   | 0.32681787758256103 | 0.4856941947250095  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (23 total) |
| 193      | 393    | 200      | 23              | 53               | 21                | 43                 | 2.441860465116279   | 0.32681787758256103 | 0.4856941947250095  | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0013559-T01-IM5-1,<br>EVT-P-0014143-T01-IM5-111,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8, ...<br>(23 total)    |
| 418      | 443    | 25       | 9               | 67               | 9                 | 55                 | None                | 0.3393946163658613  | 0.4692950509433365  | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0023994-T01-IM6-31,<br>EVT-P-0023994-T05-IM6-30,<br>EVT-P-0023994-T06-IM6-29,<br>... (9 total)   |
| 399      | 599    | 200      | 42              | 34               | 37                | 27                 | 1.9185185185185185  | 0.3541470110878887  | 0.4508164191868243  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (42 total) |
| 121      | 321    | 200      | 11              | 65               | 8                 | 56                 | 0.42857142857142855 | 0.365242647320004   | 0.43741851806441334 | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5, ...<br>(11 total)     |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value             | neg_log10_p_value   | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|---------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 143      | 343    | 200      | 12              | 64               | 9                 | 55                 | 0.4909090909090909 | 0.3901002779144654  | 0.40882374028888596 | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5, ...<br>(12 total)     |
| 159      | 359    | 200      | 13              | 63               | 10                | 54                 | 0.5555555555555556 | 0.4206035868861286  | 0.3761270279925161  | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5, ...<br>(13 total)     |
| 423      | 473    | 50       | 14              | 62               | 13                | 51                 | 2.803921568627451  | 0.4465709331321285  | 0.35010954815025525 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (14 total) |
| 442      | 492    | 50       | 14              | 62               | 13                | 51                 | 2.803921568627451  | 0.4465709331321285  | 0.35010954815025525 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (14 total) |
| 452      | 652    | 200      | 18              | 58               | 14                | 50                 | 0.56               | 0.46254663915796645 | 0.33484447025957326 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (18 total) |
| 443      | 643    | 200      | 19              | 57               | 15                | 49                 | 0.6122448979591837 | 0.48095522729154383 | 0.3178953507472748  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (19 total) |
| 417      | 467    | 50       | 22              | 54               | 20                | 44                 | 2.272727272727273  | 0.4906657002169846  | 0.3092143001343047  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (22 total) |
| 476      | 501    | 25       | 4               | 72               | 3                 | 61                 | 0.5409836065573771 | 0.504763537870964   | 0.2969120243529968  | EVT-P-0003705-T01-IM5-24,<br>EVT-P-0007755-T02-IM5-117,<br>EVT-P-0017558-T02-IM7-3,<br>EVT-P-0024231-T01-IM6-33                                                  |
| 411      | 511    | 100      | 28              | 48               | 25                | 39                 | 1.9230769230769231 | 0.5176833019050124  | 0.28593584314537324 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (28 total) |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value            | neg_log10_p_value   | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|--------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 414      | 514    | 100      | 27              | 49               | 24                | 40                 | 1.8                | 0.5213280124699622 | 0.2828889385407058  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (27 total) |
| 417      | 517    | 100      | 26              | 50               | 23                | 41                 | 1.6829268292682926 | 0.5294734121362175 | 0.27615584339868837 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (26 total) |
| 87       | 187    | 100      | 8               | 68               | 6                 | 58                 | 0.5172413793103449 | 0.6046542038537166 | 0.21849292335171544 | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0024875-T01-IM6-99,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>... (8 total)   |
| 271      | 321    | 50       | 9               | 67               | 7                 | 57                 | 0.6140350877192983 | 0.6271831003252879 | 0.20260565238732078 | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0045731-T01-IM6-35,<br>... (9 total)      |
| 243      | 343    | 100      | 10              | 66               | 8                 | 56                 | 0.7142857142857143 | 0.6537687167200023 | 0.1845758645463     | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0045731-T01-IM6-35,<br>... (10 total)     |
| 452      | 477    | 25       | 12              | 64               | 11                | 53                 | 2.2830188679245285 | 0.6777385747932327 | 0.16893779495157404 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (12 total) |
| 427      | 452    | 25       | 13              | 63               | 12                | 52                 | 2.5384615384615383 | 0.6786540902057518 | 0.168351529097454   | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (13 total) |
| 177      | 377    | 200      | 14              | 62               | 11                | 53                 | 0.6226415094339622 | 0.684604940387127  | 0.16455997114157436 | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5, ...<br>(14 total)     |
| 179      | 379    | 200      | 15              | 61               | 12                | 52                 | 0.6923076923076923 | 0.6947789745681874 | 0.15815333266474796 | EVT-P-0006172-T01-IM5-91,<br>EVT-P-0009855-T01-IM5-13,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>... (15 total)    |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value            | neg_log10_p_value   | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|--------------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 423      | 523    | 100      | 18              | 58               | 16                | 48                 | 1.6666666666666667 | 0.7198940138934029 | 0.14273143769289975 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (18 total) |
| 423      | 623    | 200      | 20              | 56               | 16                | 48                 | 0.6666666666666666 | 0.7215985127946598 | 0.14170437054058568 | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (20 total) |
| 402      | 452    | 50       | 25              | 51               | 22                | 42                 | 1.5714285714285714 | 0.740344859540053  | 0.1305659347182275  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0007050-T01-IM5-90,<br>... (25 total) |
| 416      | 616    | 200      | 28              | 48               | 23                | 41                 | 0.7853658536585366 | 0.7509447780617824 | 0.1243919983684352  | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (28 total) |
| 95       | 120    | 25       | 6               | 70               | 5                 | 59                 | 0.9322033898305084 | 1.0                | -0.0                | EVT-P-0024875-T01-IM6-99,<br>EVT-P-0042526-T02-IM6-92,<br>EVT-P-0055621-T01-IM6-97,<br>EVT-P-0055621-T02-IM6-96,<br>EVT-P-0055621-T05-IM6-95,<br>... (6 total)   |
| 358      | 383    | 25       | 4               | 72               | 4                 | 60                 | None               | 1.0                | -0.0                | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0027278-T01-IM6-45,<br>EVT-P-0042965-T01-IM6-114,<br>EVT-P-0065034-T02-IM7-115                                                |
| 405      | 430    | 25       | 12              | 64               | 10                | 54                 | 0.9259259259259259 | 1.0                | -0.0                | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0022925-T01-IM6-16,<br>EVT-P-0023994-T01-IM6-31,<br>EVT-P-0023994-T05-IM6-30,<br>... (12 total)  |
| 411      | 436    | 25       | 11              | 65               | 10                | 54                 | 2.037037037037037  | 1.0                | -0.0                | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0022925-T01-IM6-16,<br>EVT-P-0023994-T01-IM6-31,<br>EVT-P-0023994-T05-IM6-30,<br>... (11 total)  |
| 414      | 439    | 25       | 10              | 66               | 9                 | 55                 | 1.8                | 1.0                | -0.0                | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0023994-T01-IM6-31,<br>EVT-P-0023994-T05-IM6-30,<br>EVT-P-0023994-T06-IM6-29,<br>... (10 total)  |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value | neg_log10_p_value | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|---------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 417      | 442    | 25       | 9               | 67               | 8                 | 56                 | 1.5714285714285714 | 1.0     | -0.0              | EVT-P-0010086-T01-IM5-18,<br>EVT-P-0010581-T01-IM5-17,<br>EVT-P-0023994-T01-IM6-31,<br>EVT-P-0023994-T05-IM6-30,<br>EVT-P-0023994-T06-IM6-29,<br>... (9 total)   |
| 293      | 343    | 50       | 4               | 72               | 4                 | 60                 | None               | 1.0     | -0.0              | EVT-P-0045731-T01-IM6-35,<br>EVT-P-0049078-T01-IM6-103,<br>EVT-P-0049078-T04-IM6-102,<br>EVT-P-0069266-T01-IM7-112                                               |
| 309      | 359    | 50       | 4               | 72               | 4                 | 60                 | None               | 1.0     | -0.0              | EVT-P-0049078-T01-IM6-103,<br>EVT-P-0049078-T04-IM6-102,<br>EVT-P-0065034-T02-IM7-115,<br>EVT-P-0069266-T01-IM7-112                                              |
| 329      | 379    | 50       | 4               | 72               | 4                 | 60                 | None               | 1.0     | -0.0              | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0042965-T01-IM6-114,<br>EVT-P-0065034-T02-IM7-115,<br>EVT-P-0069266-T01-IM7-112                                               |
| 333      | 383    | 50       | 5               | 71               | 5                 | 59                 | None               | 1.0     | -0.0              | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0027278-T01-IM6-45,<br>EVT-P-0042965-T01-IM6-114,<br>EVT-P-0065034-T02-IM7-115,<br>EVT-P-0069266-T01-IM7-112                  |
| 451      | 501    | 50       | 16              | 60               | 14                | 50                 | 1.4                | 1.0     | -0.0              | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (16 total) |
| 259      | 359    | 100      | 11              | 65               | 9                 | 55                 | 0.8181818181818182 | 1.0     | -0.0              | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0045731-T01-IM6-35,<br>... (11 total)     |
| 277      | 377    | 100      | 12              | 64               | 10                | 54                 | 0.9259259259259259 | 1.0     | -0.0              | EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5,<br>EVT-P-0042965-T01-IM6-114,<br>... (12 total)    |
| 279      | 379    | 100      | 13              | 63               | 11                | 53                 | 1.0377358490566038 | 1.0     | -0.0              | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0038568-T01-IM6-85,<br>EVT-P-0041874-T01-IM6-5, ...<br>(13 total)     |
| 283      | 383    | 100      | 14              | 62               | 12                | 52                 | 1.1538461538461537 | 1.0     | -0.0              | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0027278-T01-IM6-45,<br>EVT-P-0038568-T01-IM6-85,<br>... (14 total)    |

| start_aa | end_aa | width_aa | n_events_inside | n_events_outside | n_positive_inside | n_positive_outside | odds_ratio         | p_value | neg_log10_p_value | event_ids_inside                                                                                                                                                 |
|----------|--------|----------|-----------------|------------------|-------------------|--------------------|--------------------|---------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 401      | 501    | 100      | 29              | 47               | 25                | 39                 | 1.2820512820512822 | 1.0     | -0.0              | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (29 total) |
| 443      | 543    | 100      | 17              | 59               | 15                | 49                 | 1.530612244897959  | 1.0     | -0.0              | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (17 total) |
| 183      | 383    | 200      | 15              | 61               | 13                | 51                 | 1.2745098039215685 | 1.0     | -0.0              | EVT-P-0009855-T01-IM5-13,<br>EVT-P-0014859-T01-IM6-9,<br>EVT-P-0014859-T03-IM6-8,<br>EVT-P-0027278-T01-IM6-45,<br>EVT-P-0038568-T01-IM6-85,<br>... (15 total)    |
| 405      | 605    | 200      | 30              | 46               | 25                | 39                 | 0.8974358974358975 | 1.0     | -0.0              | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (30 total) |
| 411      | 611    | 200      | 29              | 47               | 25                | 39                 | 1.2820512820512822 | 1.0     | -0.0              | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (29 total) |
| 414      | 614    | 200      | 28              | 48               | 24                | 40                 | 1.2                | 1.0     | -0.0              | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (28 total) |
| 418      | 618    | 200      | 27              | 49               | 23                | 41                 | 1.1219512195121952 | 1.0     | -0.0              | EVT-P-0000104-T02-IM6-118,<br>EVT-P-0003329-T02-IM6-28,<br>EVT-P-0003329-T03-IM6-26,<br>EVT-P-0003329-T04-IM6-25,<br>EVT-P-0003705-T01-IM5-24,<br>... (27 total) |

Per-event results (results.tsv)

| event_id                | sample_id         | patient_id | tumor_type     | oncogene_code | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                   | source_site2_effect_on_frame |
|-------------------------|-------------------|------------|----------------|---------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0013559-T01-IM5-1 | P-0013559-T01-IM5 |            | Glioma         | GB            | AFAP1-NTRK1   | AFAP1        | in-frame     | three_prime | 156844093          | 9               | 393                         | True                   | 393                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | AFAP1(NM_00134647) - NTRK1(NM_002529) Rearrangement : t(1;4)(q23.1;p16.1)(chr1:g.156844093::chr4:g.7852375)   Protein fusion: in frame (AFAP1-NTRK1)     | NA                           |
| EVT-P-0042270-T01-IM6-2 | P-0042270-T01-IM6 |            | Bladder Cancer | BLCA          | ANKRD36-NTRK1 | ANKRD36      | out-of-frame | three_prime | 156844593          | 11              | 418                         | True                   | 418                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ANKRD36 (NM_001164315) - NTRK1 (NM_002529) fusion: t(1;2)(q23.1;q11.2)(chr1:g.156844593::chr2:g.97835462)   Protein Fusion: out of frame {ANKRD36:NTRK1} | NA                           |

| event_id                | sample_id         | patient_id | tumor_type          | oncogene_code | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                    | source_site2_effect_on_frame |
|-------------------------|-------------------|------------|---------------------|---------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0017558-T02-IM7-3 | P-0017558-T02-IM7 |            | Ampullary Cancer    | AMPCA         | ARHGEF11-NTRK1 | ARHGEF11     | out-of-frame | three_prime | 156845502          | 13              | 501                         | True                   | 501                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ARHGEF11 (NM_198236) - NTRK1 (NM_002529) fusion: c. 32+488: ARHGEF11_c.1501+44: NTRK1in v   Protein Fusion: out of frame {ARHGEF11:NTRK1} | NA                           |
| EVT-P-0050637-T01-IM6-4 | P-0050637-T01-IM6 |            | Glioma              | GB            | BCAN-NTRK1     | BCAN         | in-frame     | three_prime | 156842373          | 8               | 284                         | True                   | 284                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | BCAN (NM_021948) - NTRK1 (NM_002529) fusion: c. 1064-525:BCAN_c.850+826:NTRK1del   Protein Fusion: in frame {BCAN:NTRK1}                  | NA                           |
| EVT-P-0041874-T01-IM6-5 | P-0041874-T01-IM6 |            | Soft Tissue Sarcoma | DDLS          | BCAN-NTRK1     | BCAN         | in-frame     | three_prime | 156843258          | 8               | 284                         | True                   | 284                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | BCAN (NM_021948) - NTRK1 (NM_002529) fusion: c. 92-224:BCAN_c.851-167: NTRK1del   Protein Fusion: in frame {BCAN:NTRK1}                   | NA                           |

| event_id                | sample_id         | patient_id | tumor_type        | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains     | source_annotation_text                                                                                                               | source_site2_effect_on_frame |
|-------------------------|-------------------|------------|-------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0055952-T01-IM6-6 | P-0055952-T01-IM6 |            | Pancreatic Cancer | PAAD          | COP1-NTRK1  | COP1         | in-frame     | three_prime | 156851106          | 17              | 736                         | True                   | 736                                          | True                               | exact match                        | disrupted     | 0.17037037037037         | True                |                       |              | Protein kinase domain | RFWD2 (NM_022457) - NTRK1 (NM_002529) rearrangement: c.2133+18462:RFWD2_c.2206-143:NTRK1inv   Protein Fusion: in frame {RFWD2:NTRK1} | NA                           |
| EVT-P-0014859-T03-IM6-8 | P-0014859-T03-IM6 |            | Pancreatic Cancer | PAAD          | CTRC-NTRK1  | CTRC         | unknown      | three_prime | 156841593          | 8               | 284                         | True                   | 284                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                       | CTRC (NM_007272) - NTRK1 (NM_002529) rearrangement: c.88:CTRC_c.850+46:NTRK1del   Protein Fusion: mid-exon {CTRC:NTRK1}              | NA                           |
| EVT-P-0014859-T01-IM6-9 | P-0014859-T01-IM6 |            | Pancreatic Cancer | PAAD          | CTRC-NTRK1  | CTRC         | unknown      | three_prime | 156841592          | 8               | 284                         | True                   | 284                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                       | CTRC (NM_007272) - NTRK1 (NM_002529) rearrangement: c.88:CTRC_c.850+45:NTRK1del   Protein fusion: mid-exon (CTRC-NTRK1)              | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type                 | oncogene_code | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                  | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|----------------------------|---------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0029326-T01-IM6-10 | P-0029326-T01-IM6 |            | Thyroid Cancer             | THPA          | DIAPH1-NTRK1 | DIAPH1       | in-frame     | three_prime | 156843860          | 9               | 393                         | True                   | 393                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | DIAPH1 (NM_005219) - NTRK1 (NM_002529) rearrangement: t(1;5)(q23.1;q31.3)(chr1:g.156843860::chr5:g.140904837)   Protein Fusion: in frame {DIAPH1:NTRK1} | NA                           |
| EVT-P-0059139-T01-IM7-11 | P-0059139-T01-IM7 |            | Pancreatic Cancer          | PAAD          | EML4-NTRK1   | EML4         | in-frame     | three_prime | 156844256          | 10              | 399                         | True                   | 399                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | EML4 (NM_019063) - NTRK1 (NM_002529) fusion: t(1;2)(q23.1;p21)(chr1:g.156844256::chr2:g.42473987)   Protein Fusion: in frame {EML4:NTRK1}               | NA                           |
| EVT-P-0037901-T01-IM6-12 | P-0037901-T01-IM6 |            | Non-Small Cell Lung Cancer | LUAD          | EPS15-NTRK1  | EPS15        | in-frame     | three_prime | 156844212          | 10              | 399                         | True                   | 399                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | EPS15 (NM_001981) - NTRK1 (NM_002529) rearrangement: c.2119+1240:EPS15_c.1195+20:NTRK1inv   Protein Fusion: in frame {EPS15:NTRK1}                      | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type     | oncotre_e_code | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intro_nic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                          | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|----------------|----------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|-------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0009855-T01-IM5-13 | P-0009855-T01-IM5 |            | Melanoma       | SKCM           | GON4L-NTRK1  | GON4L        | unknown      | three_prime | 156843711          | 8               | 379                         | False                   | 379                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | GON4L (NM_001037533) - NTRK1 (NM_002529) rearrangement : c. 4141:GON4L_c.1137:NTRK1inv   Protein fusion: mid-exon (GON4L-NTRK1) | NA                           |
| EVT-P-0050669-T01-IM6-15 | P-0050669-T01-IM6 |            | Ovarian Cancer | HGSFT          | IQGAP3-NTRK1 | IQGAP3       | unknown      | three_prime | 156844408          | 10              | 414                         | False                   | 414                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | IQGAP3 (NM_178229) - NTRK1 (NM_002529) fusion: c. 2531-131:IQGAP3_c.1241:NTRK1inv   Protein Fusion: mid-exon {IQGAP3:NTRK1}     | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type          | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                              | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|---------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0022925-T01-IM6-16 | P-0022925-T01-IM6 |            | Soft Tissue Sarcoma | SARCNOS       | LMNA-NTRK1  | LMNA         | unknown      | three_prime | 156844398          | 10              | 411                         | False                  | 411                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-3 fused to NTRK1 exons 10-17): c.513+1675:LMNA_c.1231:NTRK1del   Protein Fusion: mid-exon {LMNA:NTRK1}    | NA                           |
| EVT-P-0010581-T01-IM5-17 | P-0010581-T01-IM5 |            | Colorectal Cancer   | COAD          | LMNA-NTRK1  | LMNA         | in-frame     | three_prime | 156844556          | 11              | 418                         | True                   | 418                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 11-17): c.514-811:LMNA_c.1251+138:NTRK1del   Protein fusion: in frame (LMNA-NTRK1) | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type          | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                        | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|---------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0010086-T01-IM5-18 | P-0010086-T01-IM5 |            | Soft Tissue Sarcoma | SARCNOS       | LMNA-NTRK1  | LMNA         | in-frame     | three_prime | 156844436          | 11              | 418                         | True                   | 418                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | LMNA (NM_170707) - NTRK1 (NM_002529) rearrangement: c.514-1686:LMNA_c.1251+18:NTRK1del   Protein fusion: in frame (LMNA-NTRK1)                                | NA                           |
| EVT-P-0007739-T01-IM5-19 | P-0007739-T01-IM5 |            | Colorectal Cancer   | COAD          | LMNA-NTRK1  | LMNA         | unknown      | three_prime | 156845035          | 12              | 452                         | True                   | 452                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | LMNA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exons 1-8 fused to NTRK1 exons 12-17): c.1443:LMNA_c.1354+235:NTRK1   Protein fusion: mid-exon (LMNA-NTRK1) | NA                           |
| EVT-P-0003705-T01-IM5-24 | P-0003705-T01-IM5 |            | Appendiceal Cancer  | MAAP          | LMNA-NTRK1  | LMNA         | unknown      | three_prime | 156845433          | 12              | 492                         | False                  | 492                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | c.686:LMNA_c.1476:NTRK1del   Protein fusion: mid-exon (LMNA-NTRK1)                                                                                            | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type           | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|----------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0003329-T04-IM6-25 | P-0003329-T04-IM6 |            | Hepatobiliary Cancer | GBC           | LMNA-NTRK1  | LMNA         | unknown      | three_prime | 156845200          | 12              | 452                         | True                   | 452                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | LMNA ( NM_170707) - NTRK1 ( NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 12 - 17): c.495:LMNA_A_c.1355-112:NTRK1del   Protein Fusion: mid-exon {LMNA:NTRK1} | NA                           |
| EVT-P-0003329-T03-IM6-26 | P-0003329-T03-IM6 |            | Hepatobiliary Cancer | GBC           | LMNA-NTRK1  | LMNA         | unknown      | three_prime | 156845200          | 12              | 452                         | True                   | 452                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | LMNA ( NM_170707) - NTRK1 ( NM_002529) fusion (LMNA exons 1-2 fused to NTRK1 exons 12 - 17): c.495:LMNA_A_c.1355-112:NTRK1del   Protein Fusion: mid-exon {LMNA:NTRK1} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type           | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                   | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|----------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0066239-T01-IM7-27 | P-0066239-T01-IM7 |            | Colorectal Cancer    | COAD          | LMNA-NTRK1  | LMNA         | unknown      | three_prime | 156845170          | 12              | 452                         | True                   | 452                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1428:LMNA_c.1355-142:NTRK1del   Protein Fusion: mid-exon {LMNA:NTRK1}                                     | NA                           |
| EVT-P-0003329-T02-IM6-28 | P-0003329-T02-IM6 |            | Hepatobiliary Cancer | GBC           | LMNA-NTRK1  | LMNA         | unknown      | three_prime | 156845200          | 12              | 452                         | True                   | 452                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | LMNA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exon2 fused with NTRK1 exon12): c.495:LMNA_c.1355-112:NTRK1del   Protein Fusion: mid-exon {LMNA:NTRK1} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type    | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                             | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|---------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0023994-T06-IM6-29 | P-0023994-T06-IM6 |            | Breast Cancer | BRCA          | LMNA-NTRK1  | LMNA         | in-frame     | three_prime | 156844525          | 11              | 418                         | True                   | 418                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | LMNA (NM_170707) - NTRK1 (NM_002529) fusion: c.1699-362:LMNA_c.1251+107:NTRK1del   Protein Fusion: in frame {LMNA: NTRK1}                                          | NA                           |
| EVT-P-0023994-T05-IM6-30 | P-0023994-T05-IM6 |            | Breast Cancer | IDC           | LMNA-NTRK1  | LMNA         | in-frame     | three_prime | 156844525          | 11              | 418                         | True                   | 418                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | LMNA (NM_170707) - NTRK1 (NM_002529) Fusion (LMNA exon 10 fused with NTRK1 exon 11) : c.1699-362:LMNA_c.1251+107:NTRK1del   Protein Fusion: in frame {LMNA: NTRK1} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type             | oncotre_e_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains     | source_annotation_text                                                                                                                                                 | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|------------------------|----------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0023994-T01-IM6-31 | P-0023994-T01-IM6 |            | Breast Cancer          | BRCNO S        | LMNA-NTRK1  | LMNA         | in-frame     | three_prime | 156844525          | 11              | 418                         | True                   | 418                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                       | LMNA (NM_170707) - NTRK1 (NM_002529) fusion (LMNA exons 1-10 fused to NTRK1 exons 11-17): c.1699-362:LMNA_c.1251+107:NTRK1del   Protein Fusion: in frame {LMNA: NTRK1} | NA                           |
| EVT-P-0025031-T02-IM6-32 | P-0025031-T02-IM6 |            | Esophagogastric Cancer | ESCA           | LTAP1-NTRK1 | LTAP1        | unknown      | three_prime | 156846272          | 14              | 571                         | False                  | 571                                          | True                               | exact match                        | disrupted     | 0.7814814814814          | True                |                       |              | Protein kinase domain | C1orf43 (NM_001098616) - NTRK1 (NM_002529): c.168+114: C1orf43_c.1713: NTRK1inv   Protein Fusion: mid-exon {C1orf43: NTRK1}                                            | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type             | oncogene_code | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail                                                                                                                                                                             | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                               | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|------------------------|---------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------|---------------------|-----------------------|-----------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0024231-T01-IM6-33 | P-0024231-T01-IM6 |            | Breast Cancer          | BRCA          | MEF2D-NTRK1  | MEF2D        | in-frame     | five_prime  | 156845756          | 12              | 501                         | True                   | 500                                          | True                               | hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement | lost          | 0.0                      | False               |                       | Protein kinase domain |                   | NTRK1 (NM_002529) - MEF2D (NM_005920) rearrangement: c.1502-116:NTRK1_c.1006+1908:MEF2Binv   Protein Fusion: in frame {NTRK1: MEF2D} | NA                           |
| EVT-P-0045731-T01-IM6-35 | P-0045731-T01-IM6 |            | Esophagogastric Cancer | ESCA          | NOS1AP-NTRK1 | NOS1AP       | unknown      | three_prime | 156843455          | 8               | 294                         | False                  | 294                                          | True                               | exact match                                                                                                                                                                                                    | retained      | 1.0                      | False               | Protein kinase domain |                       |                   | NOS1AP (NM_014697) - NTRK1 (NM_002529) fusion: c.1106-363:NOS1AP_c.881:NTRK1dup   Protein Fusion: mid-exon {NOS1AP:NTRK1}            | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type                 | oncogene_code | fusion_name    | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                      | source_site_effect_on_frame |
|--------------------------|-------------------|------------|----------------------------|---------------|----------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| EVT-P-0026354-T01-IM6-36 | P-0026354-T01-IM6 |            | Melanoma                   | MUP           | NTRK1-ARHGEF11 | ARHGEF11     | in-frame     | three_prime | 156845184          | 12              | 452                         | True                   | 452                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ARHGEF11 (NM_198236) - NTRK1 (NM_002529) rearrangement: c.4631-285:ARHGEF11_c.1355-128:NTRK1inv   Protein Fusion: in frame {ARHGEF11:NTRK1} | NA                          |
| EVT-P-0035466-T01-IM6-37 | P-0035466-T01-IM6 |            | Non-Small Cell Lung Cancer | LUAD          | NTRK1-ATP1A2   | ATP1A2       | unknown      | three_prime | 156844714          | 11              | 423                         | False                  | 423                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ATP1A2 (NM_000702) - NTRK1 (NM_002529) rearrangement: c.2563+2:ATP1A2_c.1268:NTRK1dup   Protein Fusion: mid-exon {ATP1A2:NTRK1}             | NA                          |

| event_id                 | sample_id         | patient_id | tumor_type                 | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail                                                                                                                                                                             | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                | source_site2_effect_on_frame |
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| EVT-P-0058090-T04-IM7-38 | P-0058090-T04-IM7 |            | Non-Small Cell Lung Cancer | LUAD          | NTRK1-CADM3 | CADM3        | out-of-frame | five_prime  | 156836298          | 3               | 120                         | True                   | 119                                          | True                               | hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement | lost          | 0.0                      | False               |                       | Protein kinase domain |                   | NTRK1 (NM_002529) - CADM3 (NM_021189) rearrangement: c.360-404: NTRK1_c.88+5312:CADM3del   Protein Fusion: out of frame {NTRK1:CADM3} | NA                           |
| EVT-P-0021262-T01-IM6-41 | P-0021262-T01-IM6 |            | Non-Small Cell Lung Cancer | LUAD          | NTRK1-F11R  | F11R         | in-frame     | three_prime | 156844288          | 10              | 399                         | True                   | 399                                          | True                               | exact match                                                                                                                                                                                                    | retained      | 1.0                      | False               | Protein kinase domain |                       |                   | F11R (NM_016946) - NTRK1 (NM_002529) rearrangement: c.389-101:F11R_c.1196-75:NTKR1inv   Protein Fusion: in frame {F11R:NTRK1}         | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type                | oncogene_code | fusion_name   | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                            | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|---------------------------|---------------|---------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0022467-T01-IM6-42 | P-0022467-T01-IM6 |            | Thyroid Cancer            | THAP          | NTRK1-IRF2BP2 | IRF2BP2      | in-frame     | three_prime | 156844129          | 9               | 393                         | True                   | 393                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | IRF2BP2 (NM_001077397) - NTRK1 (NM_002529) fusion (IRF2BP2 exon 1 fused with NTRK1 exons 9-17): c. 1001-193:IRF2BP2_c.1178-46:NTRK1inv   Protein Fusion: in frame (IRF2BP2:NTRK1) | NA                           |
| EVT-P-0010628-T01-IM5-43 | P-0010628-T01-IM5 |            | Cancer of Unknown Primary | PDC           | NTRK1-IRF2BP2 | IRF2BP2      | unknown      | three_prime | 156844224          | 10              | 399                         | True                   | 399                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | IRF2BP2 (NM_001077397) - NTRK1 (NM_002529) rearrangement : c.948:IRF2BP2_1195+32:NTRK1inv   Protein fusion: mid-exon (IRF2BP2-NTRK1)                                              | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type                 | oncogene_code | fusion_name   | partner_gene | frameshift | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                | source_site2_effect_on_frame |
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| EVT-P-0010628-T03-IM6-44 | P-0010628-T03-IM6 |            | Non-Small Cell Lung Cancer | LUAD          | NTRK1-IRF2BP2 | IRF2BP2      | unknown    | three_prime | 156844223          | 10              | 399                         | True                   | 399                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | IRF2BP2 (NM_001077397) - NTRK1 (NM_002529) rearrangement: c.950:IRF2BP2_c.1195+31:NTRK1inv   Protein Fusion: mid-exon {IRF2BP2:NTRK1} | NA                           |
| EVT-P-0027278-T01-IM6-45 | P-0027278-T01-IM6 |            | Miscellaneous Brain Tumor  | PRNET         | NTRK1-KIF21B  | KIF21B       | unknown    | three_prime | 156843723          | 8               | 383                         | False                  | 383                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | KIF21B (NM_001252100) - NTRK1 (NM_002529) rearrangement: c.2077+601:KIF21B_c.1149:NTRK1inv   Protein Fusion: mid-exon {KIF21B:NTRK1}  | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type          | oncogene_code | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail                                                                                                                                                                            | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                                     | source_site2_effect_on_frame |
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| EVT-P-0056197-T01-IM6-46 | P-0056197-T01-IM6 |            | Ovarian Cancer      | HGSOC         | NTRK1-NELFCD | NELFCD       | out-of-frame | five_prime  | 156843148          | 7               | 284                         | True                   | 283                                           | True                                | hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement | lost          | 0.0                      | False               |                       | Protein kinase domain |                   | NTRK1 (NM_002529) - NELFCD (NM_198976) rearrangement: t(1;20)(q23.1;q13.32)(chr1:g.156843148::chr20:g.57565762)   Protein Fusion: out of frame {NTRK1:NELFCD}              | NA                           |
| EVT-P-0027277-T01-IM6-79 | P-0027277-T01-IM6 |            | Soft Tissue Sarcoma | ANGS          | NTRK1-PEAR1  | PEAR1        | in-frame     | three_prime | 156844219          | 10              | 399                         | True                   | 399                                           | True                                | exact match                                                                                                                                                                                                    | retained      | 1.0                      | False               | Protein kinase domain |                       |                   | PEAR1 (NM_001080471) - NTRK1 (NM_002529) fusion (PEAR1 exons 1-15 fused to NTRK1 exons 9-17): c.1951+466:PEAR1_c.1195+27:NTKR1dup   Protein Fusion: in frame {PEAR1:NTRK1} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type           | oncogene_code | fusion_name       | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                              | source_site2_effect_on_frame |
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| EVT-P-0029892-T05-IM6-80 | P-0029892-T05-IM6 |            | Hepatobiliary Cancer | CHOL          | NTRK1-<br>PLEKHA6 | PLEKHA6      | in-frame     | three_prime | 156844239          | 10              | 399                         | True                   | 399                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | PLEKHA6 (NM_014935) - NTRK1 (NM_002529) fusion: c. 2032+105:PLEKHA6_c.1195+47: NTRK1in v   Protein Fusion: in frame {PLEKHA6:NTRK1} | NA                           |
| EVT-P-0029892-T04-IM6-81 | P-0029892-T04-IM6 |            | Hepatobiliary Cancer | IHCH          | NTRK1-<br>PLEKHA6 | PLEKHA6      | in-frame     | three_prime | 156844239          | 10              | 399                         | True                   | 399                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | PLEKHA6 (NM_014935) - NTRK1 (NM_002529) fusion: c. 2032+105:PLEKHA6_c.1195+47: NTRK1in v   Protein Fusion: in frame {PLEKHA6:NTRK1} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type           | oncogene_code | fusion_name       | partner_gene | frameshift   | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                              | source_site2_effect_on_frame |
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| EVT-P-0029892-T03-IM6-82 | P-0029892-T03-IM6 |            | Hepatobiliary Cancer | IHCH          | NTRK1-<br>PLEKHA6 | PLEKHA6      | in-frame     | three_prime | 156844239          | 10              | 399                         | True                   | 399                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |                       |                   | PLEKHA6 (NM_014935) - NTRK1 (NM_002529) fusion: c. 2032+105;PLEKHA6_c.1195+47: NTRK1in v   Protein Fusion: in frame {PLEKHA6:NTRK1} | NA                           |
| EVT-P-0038819-T01-IM6-83 | P-0038819-T01-IM6 |            | Pancreatic Cancer    | PAAD          | NTRK1-<br>RAB25   | RAB25        | out-of-frame | five_prime  | 156844458          | 10              | 417                         | True                   | 417                                          | True                               | exact match                        | lost          | 0.0                      | False               |                       | Protein kinase domain |                   | NTRK1 (NM_002529) - RAB25 (NM_020387) Rearrangement :   Protein Fusion: out of frame {NTRK1:RAB25}                                  | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type          | oncogene_code | fusion_name     | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail                                                                                                                                                                             | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                               | source_site2_effect_on_frame |
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| EVT-P-0015004-T01-IM6-84 | P-0015004-T01-IM6 |            | Soft Tissue Sarcoma | WDLS          | NTRK1-RABGA P1L | RABGA P1L    | out-of-frame | five_prime  | 156844972          | 11              | 452                         | True                   | 451                                          | True                               | hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement | lost          | 0.0                      | False               |                  | Protein kinase domain |                   | NTRK1 (NM_002529) - RABGA P1L (NM_014857) rearrangement: c.1354+172:NTRK1_c.2340+50594:RABGA P1Ldel   Protein fusion: out of frame (NTRK1-RABGA P1L) | NA                           |
| EVT-P-0038568-T01-IM6-85 | P-0038568-T01-IM6 |            | Prostate Cancer     | PRAD          | NTRK1-SEMA4A    | SEMA4A       | out-of-frame | five_prime  | 156841963          | 7               | 284                         | True                   | 283                                          | True                               | hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement | lost          | 0.0                      | False               |                  | Protein kinase domain |                   | NTRK1 (NM_002529) - SEMA4A (NM_001193300) rearrangement: c.850+416:NTRK1_c.301-270:SEMA4Adup   Protein Fusion: out of frame {NTRK1:S SEMA4A}         | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type          | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                        | source_site2_effect_on_frame |
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| EVT-P-0048896-T01-IM6-86 | P-0048896-T01-IM6 |            | Glioma              | GB            | NTRK1-SMYD2 | SMYD2        | unknown      | five_prime  | 156844382          | 10              | 405                         | False                  | 405                                          | True                               | exact match                        | lost          | 0.0                      | False               |                       | Protein kinase domain |                   | NTRK1 (NM_002529) - SMYD2 (NM_020197) rearrangement: c.1215:NTRK1_c.602+876:SMYD2del   Protein Fusion: mid-exon {NTRK1:SMYD2} | NA                           |
| EVT-P-0023987-T01-IM6-88 | P-0023987-T01-IM6 |            | Soft Tissue Sarcoma | SARCNOS       | NTRK1-TPR   | TPR          | unknown      | three_prime | 156844285          | 10              | 399                         | True                   | 399                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |                       |                   | TPR (NM_003292) - NTRK1 (NM_002529) rearrangement: c.2902:TPR_c.1196-78:NTRK1inv   Protein Fusion: mid-exon {TPR:NTRK1}       | NA                           |
| EVT-P-0024549-T03-IM6-89 | P-0024549-T03-IM6 |            | Thyroid Cancer      | THPA          | NTRK1-TPR   | TPR          | in-frame     | three_prime | 156844290          | 10              | 399                         | True                   | 399                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |                       |                   | TPR (NM_003292) - NTRK1 (NM_002529) fusion: c.2776+651_c.1196-73inv   Protein Fusion: in frame {TPR:NTRK1}                    | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type                 | oncogene_code | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail                                                                                                                                                                            | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                 | source_site2_effect_on_frame |
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| EVT-P-0007050-T01-IM5-90 | P-0007050-T01-IM5 |            | Non-Small Cell Lung Cancer | LUAD          | NTRK1-VANGL2 | VANGL2       | unknown      | three_prime | 156845141          | 12              | 452                         | True                   | 452                                           | True                                | exact match                                                                                                                                                                                                    | retained      | 1.0                      | False               | Protein kinase domain |                       |                   | VANGL2 (NM_020335) - NTRK1 (NM_002529) rearrangement : c.-191+4456:VANGL2_c.1355-171:NTRK1dup   Transcript fusion (VANGL2-NTRK1)       | NA                           |
| EVT-P-0006172-T01-IM5-91 | P-0006172-T01-IM5 |            | Non-Small Cell Lung Cancer | LUAD          | P2RY8-NTRK1  | P2RY8        | unknown      | five_prime  | 156838012          | 5               | 182                         | False                  | 181                                           | True                                | hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement | lost          | 0.0                      | False               |                       | Protein kinase domain |                   | P2RY8 (NM_178129) - NTRK1 (NM_002529) rearrangement: t(X;1)(chrX:g.1598513::chr1:g.156838012)   Protein fusion: mid-exon (NTRK1-P2RY8) | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type             | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                | source_site2_effect_on_frame |
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| EVT-P-0042526-T02-IM6-92 | P-0042526-T02-IM6 |            | Small Cell Lung Cancer | CSCLC         | PRCC-NTRK1  | PRCC         | unknown      | three_prime | 156834591          | 3               | 120                         | False                  | 120                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | PRCC (NM_005973) - NTRK1 (NM_002529) fusion: c.1323+10:PRCC_c.359:NTRK1del   Protein Fusion: mid-exon {PRCC:NTRK1}    | NA                           |
| EVT-P-004467-T01-IM6-93  | P-004467-T01-IM6  |            | Breast Cancer          | IDC           | SCP2-NTRK1  | SCP2         | unknown      | three_prime | 156844773          | 11              | 443                         | False                  | 443                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | SCP2 (NM_002979) - NTRK1 (NM_002529) fusion: c.1338+1979:SCP2_c.1327:NTRK1del   Protein Fusion: mid-exon {SCP2:NTRK1} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type          | oncogene_code | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail                                                                                                                                                                             | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains          | disrupted_domains | source_annotation_text                                                                                                                                     | source_site2_effect_on_frame |
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| EVT-P-003662-T01-IM6-94  | P-003662-T01-IM6  |            | Soft Tissue Sarcoma | DDLS          | SHPRH-NTRK1  | SHPRH        | out-of-frame | five_prime  | 156831180          | 1               | 71                          | True                   | 70                                           | True                               | hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement | lost          | 0.0                      | False               |                       | Protein kinase domain |                   | NTRK1 (NM_002529) - SHPRH (NM_001042683) rearrangement: t(1;6)(q23;q24.3)(chr1:g.156831180::chr6:g.146225530)   Protein Fusion: out of frame {NTRK1:SHPRH} | NA                           |
| EVT-P-0055621-T05-IM6-95 | P-0055621-T05-IM6 |            | Bladder Cancer      | BLCA          | SLAMF6-NTRK1 | SLAMF6       | out-of-frame | three_prime | 156836411          | 4               | 120                         | True                   | 120                                          | True                               | exact match                                                                                                                                                                                                    | retained      | 1.0                      | False               | Protein kinase domain |                       |                   | SLAMF6 (NM_052931) - NTRK1 (NM_002529) fusion: c.50-1837:SLAMF6_c.360-291:NTRK1inv   Protein Fusion: out of frame {SLAMF6:NTRK1}                           | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type     | oncogene_code | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_crosscheck_protein_position | fusion_annotation_crosscheck_agrees | fusion_annotation_crosscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                    | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|----------------|---------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|-----------------------------------------------|-------------------------------------|-------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0055621-T02-IM6-96 | P-0055621-T02-IM6 |            | Bladder Cancer | UTUC          | SLAMF6-NTRK1 | SLAMF6       | out-of-frame | three_prime | 156836411          | 4               | 120                         | True                   | 120                                           | True                                | exact match                         | retained      | 1.0                      | False               | Protein kinase domain |              |                   | SLAMF6 (NM_052931) - NTRK1 (NM_002529) rearrangement: c.50-1839:SLAMF6_c.360-291: NTRK1inv   Protein Fusion: out of frame {SLAMF6: NTRK1} | NA                           |
| EVT-P-0055621-T01-IM6-97 | P-0055621-T01-IM6 |            | Bladder Cancer | UTUC          | SLAMF6-NTRK1 | SLAMF6       | out-of-frame | three_prime | 156836411          | 4               | 120                         | True                   | 120                                           | True                                | exact match                         | retained      | 1.0                      | False               | Protein kinase domain |              |                   | SLAMF6 (NM_052931) - NTRK1 (NM_002529) rearrangement: c.50-1839:SLAMF6_c.360-291: NTRK1inv   Protein Fusion: out of frame {SLAMF6: NTRK1} | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type             | oncogene_code | fusion_name | partner_gene | frameshift | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail                                                                                                                                                                             | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains     | source_annotation_text                                                                                                                    | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|------------------------|---------------|-------------|--------------|------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0022936-T01-IM6-98 | P-0022936-T01-IM6 |            | Esophagogastric Cancer | EGC           | STK11-NTRK1 | STK11        | unknown    | five_prime  | 156848955          | 15              | 616                         | False                  | 615                                          | True                               | hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement | disrupted     | 0.388888888888889        | True                |                       |              | Protein kinase domain | NTRK1 (NM_002529) - STK11 (NM_000455) rearrangement: t(1;19)(;)(chr1:g.56848955::chr19:g.204186)   Protein Fusion: mid-exon {NTRK1:STK11} | NA                           |
| EVT-P-0024875-T01-IM6-99 | P-0024875-T01-IM6 |            | Soft Tissue Sarcoma    | SARCNOS       | TAF12-NTRK1 | TAF12        | unknown    | three_prime | 156836548          | 4               | 120                         | True                   | 120                                          | True                               | exact match                                                                                                                                                                                                    | retained      | 1.0                      | False               | Protein kinase domain |              |                       | NTRK1 (NM_002529) rearrangement: t(1;12)(q23.1;q14.1)(chr1:g.156836548::chr12:g.62314748)   Transcript Fusion {FAM19A2:NTRK1}             | NA                           |

| event_id                  | sample_id         | patient_id | tumor_type          | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|---------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0069757-T01-IM7-100 | P-0069757-T01-IM7 |            | Soft Tissue Sarcoma | DDLS          | TARS2-NTRK1 | TARS2        | unknown      | three_prime | 156838028          | 5               | 187                         | False                  | 187                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TARS2 (NM_025150) - NTRK1 (NM_002529) fusion: c. 264-188: TARS2_c.561:NTRK1del   Protein Fusion: mid-exon {TARS2:NTRK1}                               | NA                           |
| EVT-P-0050767-T01-IM6-101 | P-0050767-T01-IM6 |            | Thyroid Cancer      | THPA          | TPM3-NTRK1  | TPM3         | in-frame     | three_prime | 156845252          | 12              | 452                         | True                   | 452                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TPM3(NM_153469) - NTRK1 (NM_002529) Fusion(TPM3 exon 12 with NTRK1 exon12): c.665-654:TPM3_c.1355-60:NTRK1inv   Protein Fusion: in frame {TPM3:NTRK1} | NA                           |
| EVT-P-0049078-T04-IM6-102 | P-0049078-T04-IM6 |            | Colorectal Cancer   | COAD          | TPM3-NTRK1  | TPM3         | unknown      | three_prime | 156843537          | 8               | 321                         | False                  | 321                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TPM3 (NM_001278190) - NTRK1 (NM_002529) fusion: c. 395-6179:TPM3_c.963:NTRK1inv   Protein Fusion: mid-exon {TPM3:NTRK1}                               | NA                           |

| event_id                  | sample_id         | patient_id | tumor_type        | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                  | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|-------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0049078-T01-IM6-103 | P-0049078-T01-IM6 |            | Colorectal Cancer | COAD          | TPM3-NTRK1  | TPM3         | unknown      | three_prime | 156843537          | 8               | 321                         | False                  | 321                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TPM3 (NM_001278190) - NTRK1 (NM_002529) fusion: c.395-6179:TPM3_c.963:NTRK1inv   Protein Fusion: mid-exon {TPM3:NTRK1}  | NA                           |
| EVT-P-0046877-T02-IM6-104 | P-0046877-T02-IM6 |            | Thyroid Cancer    | THPA          | TPM3-NTRK1  | TPM3         | in-frame     | three_prime | 156843811          | 9               | 393                         | True                   | 393                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.776-357:TPM3_c.1177+60:NTRK1inv   Protein Fusion: in frame {TPM3:NTRK1}  | NA                           |
| EVT-P-0026375-T04-IM6-105 | P-0026375-T04-IM6 |            | Thyroid Cancer    | THAP          | TPM3-NTRK1  | TPM3         | in-frame     | three_prime | 156844046          | 9               | 393                         | True                   | 393                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c.775+355:TPM3_c.1178-129:NTRK1inv   Protein Fusion: in frame {TPM3:NTRK1} | NA                           |

| event_id                  | sample_id         | patient_id | tumor_type          | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                             | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|---------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0026375-T01-IM6-106 | P-0026375-T01-IM6 |            | Thyroid Cancer      | THPA          | TPM3-NTRK1  | TPM3         | in-frame     | three_prime | 156844046          | 9               | 393                         | True                   | 393                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 9-17): c.775+355:TPM3_c.1178-129:NTRK1inv   Protein Fusion: in frame {TPM3:NTRK1} | NA                           |
| EVT-P-0025815-T02-IM6-107 | P-0025815-T02-IM6 |            | Soft Tissue Sarcoma | SARCNOS       | TPM3-NTRK1  | TPM3         | in-frame     | three_prime | 156844230          | 10              | 399                         | True                   | 399                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-6 with NTRK1 exons 10-17): c.*6032:TPM3_c.1195+38:NTRK1inv   Protein Fusion: in frame {TPM3:NTRK1}       | NA                           |

| event_id                  | sample_id         | patient_id | tumor_type          | oncotre_e_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                   | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|---------------------|----------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0021467-T01-IM6-108 | P-0021467-T01-IM6 |            | Glioma              | GB             | TPM3-NTRK1  | TPM3         | in-frame     | three_prime | 156844038          | 9               | 393                         | True                   | 393                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-6 fused in-frame to NTRK1 exons 9-17): c.*652:TPM3_c.1178-137:NTRK1inv   Protein Fusion: in frame {TPM3:NTRK1} | NA                           |
| EVT-P-0021170-T03-IM6-109 | P-0021170-T03-IM6 |            | Soft Tissue Sarcoma | SARCNOS        | TPM3-NTRK1  | TPM3         | in-frame     | three_prime | 156845218          | 12              | 452                         | True                   | 452                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 12-17): c.*775+235:TPM3_c.1355-94:NTRK1inv   Protein Fusion: in frame {TPM3:NTRK1}      | NA                           |

| event_id                  | sample_id         | patient_id | tumor_type          | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                                      | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|---------------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0021170-T02-IM6-110 | P-0021170-T02-IM6 |            | Soft Tissue Sarcoma | SARCN OS      | TPM3-NTRK1  | TPM3         | in-frame     | three_prime | 156845218          | 12              | 452                         | True                   | 452                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-8 fused to NTRK1 exons 12-17): c.1355-94:NTRK1inv   Protein Fusion: in frame {TPM3:NTRK1}                         | NA                           |
| EVT-P-0014143-T01-IM5-111 | P-0014143-T01-IM5 |            | Colorectal Cancer   | COAD          | TPM3-NTRK1  | TPM3         | in-frame     | three_prime | 156844002          | 9               | 393                         | True                   | 393                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TPM3 (NM_152263) - NTRK1 (NM_002529) fusion (TPM3 exons 1-10 fused in-frame to NTRK1 exons 9-17): c.*1568:TPM3_c.1178-173: NTRK1inv   Protein fusion: in frame (TPM3-NTRK1) | NA                           |

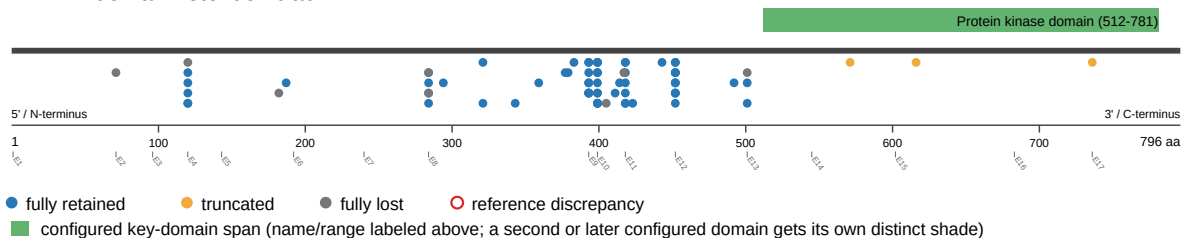
| event_id                  | sample_id         | patient_id | tumor_type     | oncogene_code | fusion_name | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                | source_site2_effect_on_frame |
|---------------------------|-------------------|------------|----------------|---------------|-------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-----------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0069266-T01-IM7-112 | P-0069266-T01-IM7 |            | Thyroid Cancer | THPA          | TPM3-NTRK1  | TPM3         | unknown      | three_prime | 156843601          | 8               | 343                         | False                  | 343                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TPM3 (NM_152263) - NTRK1 (NM_002529) fusion: c. 854+282:TPM3_c.1027:NTRK1inv   Protein Fusion: mid-exon {TPM3:NTRK1}  | NA                           |
| EVT-P-0049951-T01-IM6-113 | P-0049951-T01-IM6 |            | Thyroid Cancer | THPA          | TPR-NTRK1   | TPR          | in-frame     | three_prime | 156844660          | 11              | 418                         | True                   | 418                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TPR (NM_003292) - NTRK1 (NM_002529) fusion: c. 2610+187:TPR_c.1252-38:NTRK1inv   Protein Fusion: in frame {TPR:NTRK1} | NA                           |
| EVT-P-0042965-T01-IM6-114 | P-0042965-T01-IM6 |            | Thyroid Cancer | THPA          | TPR-NTRK1   | TPR          | unknown      | three_prime | 156843703          | 8               | 377                         | False                  | 377                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TPR (NM_003292) - NTRK1 (NM_002529) fusion: c. 2776+531:TPR_c.1129:NTRK1inv   Protein Fusion: mid-exon {TPR:NTRK1}    | NA                           |

| event_id                  | sample_id          | patient_id | tumor_type        | oncogene_code | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                                      | source_site2_effect_on_frame |
|---------------------------|--------------------|------------|-------------------|---------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-0065034-T02-IM7-115 | P-0065034-T02-I M7 |            | Colorectal Cancer | COAD          | TPR-NTRK1    | TPR          | unknown      | three_prime | 156843651          | 8               | 359                         | False                  | 359                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TPR (NM_003292) - NTRK1 (NM_002529) fusion: c. 2777-512:TPR_c.1077:NTRK1inv   Protein Fusion: mid-exon {TPR: NTRK1}                                         | NA                           |
| EVT-P-0016347-T01-IM6-116 | P-0016347-T01-I M6 |            | Melanoma          | SKCM          | TRIM63-NTRK1 | TRIM63       | in-frame     | three_prime | 156844048          | 9               | 393                         | True                   | 393                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TRIM63 (NM_032588) - NTRK1 (NM_002529) Rearrangement : c. 1051+386:TRIM63_c.1178-127:NTRK1inv   Protein Fusion: in frame {TRIM63 :NTRK1}                    | NA                           |
| EVT-P-0007755-T02-IM5-117 | P-0007755-T02-I M5 |            | Bladder Cancer    | BLCA          | TRPM8-NTRK1  | TRPM8        | out-of-frame | three_prime | 156845664          | 13              | 501                         | True                   | 501                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | TRPM8 (NM_024080) - NTRK1 (NM_002529) rearrangement: t(1;2)(q23.1;q 27.1)(chr 1:g.156845664::chr2:g.234866124)   Protein fusion: out of frame (TRPM8-NTRK1) | NA                           |

| event_id                 | sample_id         | patient_id | tumor_type    | oncogene_code | fusion_name  | partner_gene | frame_status | target_role | breakpoint_genomic | breakpoint_exon | breakpoint_protein_position | is_intronic_breakpoint | fusion_annotation_croscheck_protein_position | fusion_annotation_croscheck_agrees | fusion_annotation_croscheck_detail | domain_status | domain_retained_fraction | domain_is_truncated | retained_domains      | lost_domains | disrupted_domains | source_annotation_text                                                                                                                | source_site2_effect_on_frame |
|--------------------------|-------------------|------------|---------------|---------------|--------------|--------------|--------------|-------------|--------------------|-----------------|-----------------------------|------------------------|----------------------------------------------|------------------------------------|------------------------------------|---------------|--------------------------|---------------------|-----------------------|--------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| EVT-P-000104-T02-IM6-118 | P-0000104-T02-IM6 |            | Breast Cancer | ILC           | ZBTB7B-NTRK1 | ZBTB7B       | unknown      | three_prime | 156845173          | 12              | 452                         | True                   | 452                                          | True                               | exact match                        | retained      | 1.0                      | False               | Protein kinase domain |              |                   | ZBTB7B (NM_001252406) - NTRK1 (NM_002529) Rearrangement : c.944:ZBTB7B_c.1355-139: NTRK1del   Protein fusion: mid-exon (ZBTB7B-NTRK1) | NA                           |

## domain retention outliers

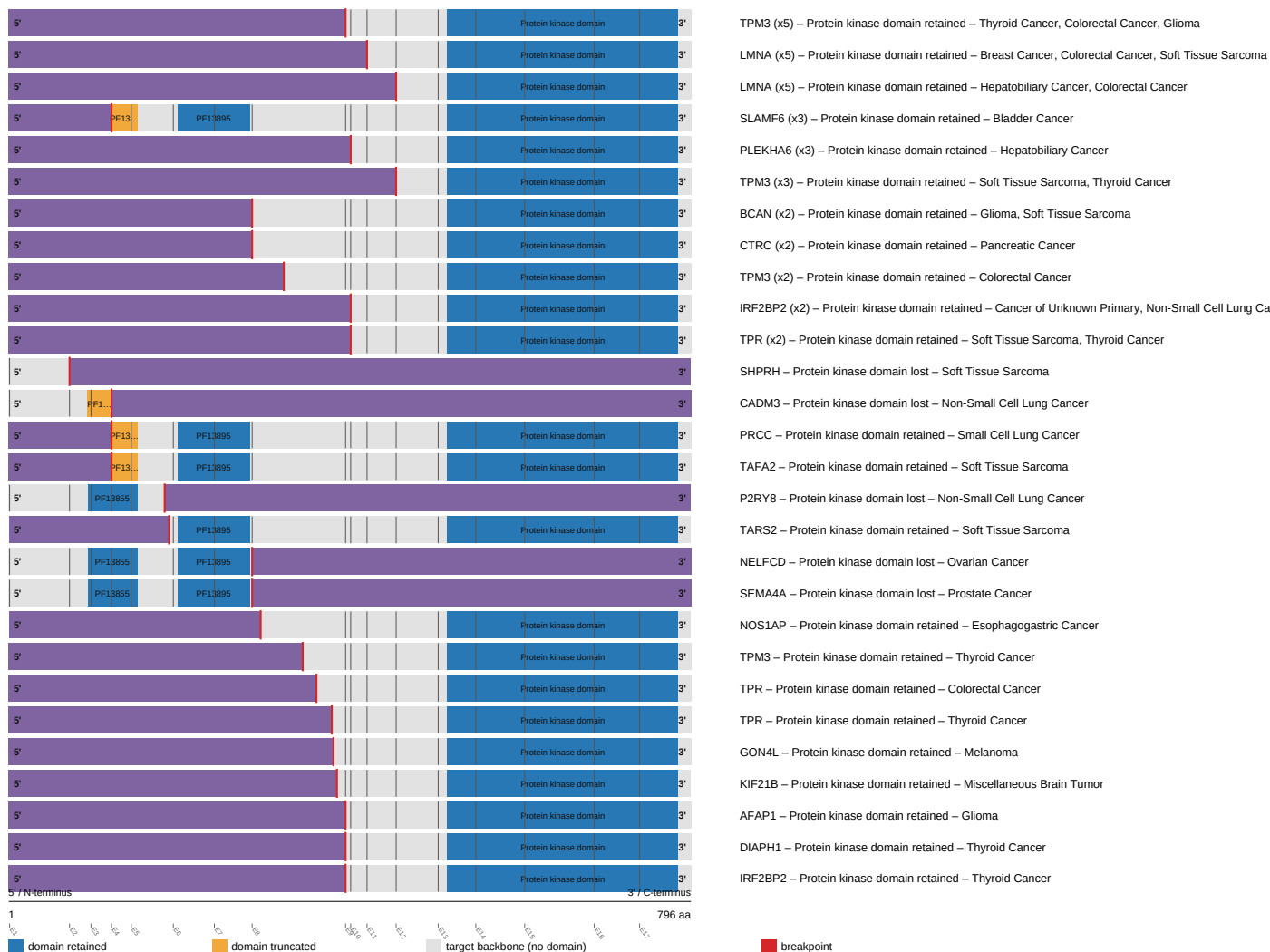
NTRK1 domain-retention track



## fusion schematic

### NTRK1 fusion-transcript schematic

Fusion order varies by row; follow each row's 5' and 3' labels (NTRK1 portion is domain-colored, exon ticks). Partner-gene blocks share one color; row labels identify each partner.



Partner-block colors are decorative; row labels identify partners.

Showing the top 28 of 53 partner/breakpoint groups by recurrence: see results.tsv for the complete list.

# intragenic deletion schematic

